



**FACULTY OF INFORMATION TECHNOLOGY AND COMMUNICATION  
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**DEPARTMENT OF INFORMATION TECHNOLOGY STUDIES**

**UNDERGRADUATE**

**WORK**

**DESIGN AND DEVELOPMENT OF A WEB BASED APPLICATION FOR  
MONITORING ENERGY EXPENDITURE AND DAILY CALORIE NEEDS**

**GROUP 42**

**BY**

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## **CANDIDATES' DECLARATION**

We hereby declare that this project work titled “Design and Development of an AI-Powered Web Application for Monitoring Energy Expenditure and Daily Caloric Needs” is the result of our own original research and development efforts carried out under the approved academic guidelines of the University. We affirm that this work has not been submitted, either wholly or in part, for any other degree, diploma, or academic qualification in this or any other institution. All sources of information, ideas, and materials obtained from published and unpublished works have been duly acknowledged and properly referenced in accordance with accepted academic standards. We further declare that any assistance received in the course of this project has been appropriately acknowledged, and that this work represents an honest and accurate account of the research conducted. We take full responsibility for any errors or omissions contained herein.

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## **SUPERVISOR’S DECLARATION**

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Supervisor’s Signature.....

Date.....

## **DEDICATION**

This project is dedicated to God Almighty for His guidance, strength, and wisdom throughout the course of this study. We also dedicate this work to our parents and families for their unwavering support, encouragement, and sacrifices, which made our academic journey possible. Special appreciation goes to our lecturers and mentors whose guidance, motivation, and academic input contributed significantly to the successful completion of this project. Finally, this work is dedicated to all individuals striving to improve their health and well-being through informed lifestyle choices, for whom this project was developed.

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## ABSTRACT

As digital health tools gain popularity, emphasis has shifted toward fitness trackers providing personalized feedback on physical activity and diet. Current calorie-tracking apps often use general formulas that overlook metabolic variations, resulting in inaccurate energy expenditure estimates and limited weight management value. This study introduces CalorieAi, a web application using AI to estimate activity-related energy expenditure and calculate tailored calorie targets for weight loss, maintenance, or gain. By integrating metabolic equivalent of task (MET) values with the Mifflin–St Jeor equation, the application offers precise energy needs and expenditure estimates. A hybrid AI approach refines these recommendations based on individual characteristics and activity patterns.

Furthermore, the system integrates advanced Generative AI (Google Gemini API) for automated, image-based food logging via computer vision (Esteva et al., 2019), alongside dynamic gamification and predictive weight-change analytics to sustain engagement (Lyons et al., 2014). Development followed a Design and Development Research (DDR) methodology aligned with Agile principles, utilizing iterative planning and evaluation cycles. Usability testing, questionnaires, and observations were conducted with students at the University of Professional Studies, Accra.

CalorieAi is built using HTML5, CSS3, and JavaScript(ES6+), it operates as a serverless application uses Firebase for backend services, including Authentication for security and Cloud Firestore for real-time data synchronization. Results show that AI-enhanced models provide more accurate calorie estimates than conventional applications. This research demonstrates that tailored web-based tools can boost user engagement and informed choices, benefiting individuals, fitness professionals, and organizations in health program development

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# CHAPTER ONE

## GENERAL INTRODUCTION

### 1.1 Introduction

Greater reliance on digital technologies has significantly changed how individuals track, plan, and manage health and fitness activities. In recent years, web-based and mobile fitness applications have become widely used tools for monitoring physical activity, estimating calorie expenditure, and supporting body weight management efforts (Krebs & Duncan, 2015). These technologies aim to enhance health-related decision-making by providing timely and easily accessible fitness data. However, the effectiveness of such systems largely depends on the accuracy of their calculations, the relevance of their recommendations, and their ability to adapt to individual user characteristics (Lyons et al., 2014).

Monitoring energy expenditure and estimating daily caloric requirements are central components of effective fitness planning and weight management. Energy expenditure varies considerably among individuals due to factors such as age, biological sex, body composition, activity intensity, and lifestyle patterns (Frayn, 2010; Westerterp, 2013). Despite the availability of numerous calorie-tracking applications, many existing systems continue to rely on generalized assumptions and static metabolic formulas that do not adequately reflect individual differences. As a result, calorie estimates produced by these systems may be inaccurate, which can reduce user confidence and long-term engagement with the application (Li & Mensah, 2023).

Inaccurate calorie estimation undermines the practical usefulness of fitness applications, particularly for users seeking to lose, maintain, or gain weight. Studies indicate that misleading feedback from digital health tools can negatively influence user motivation and adherence to

fitness goals (Krebs & Duncan, 2015). These limitations highlight the need for more intelligent and personalized approaches to energy expenditure and caloric requirement estimation.

Recent advances in artificial intelligence (AI) have introduced new opportunities to improve the accuracy and responsiveness of fitness monitoring systems. AI-powered applications are capable of analyzing user-specific data and applying intelligent logic to generate more precise estimates of calories burned during physical activity as well as daily caloric needs (Chen & Zhang, 2020; Esteva et al., 2019). When integrated with web technologies, such systems offer additional benefits, including cross-device accessibility, scalability, and ease of use without reliance on specialized hardware or wearable devices (Fielding & Taylor, 2017).

This study therefore focuses on the design and development of an AI-powered web application, CalorieAi, for monitoring energy expenditure and estimating daily caloric requirements. The proposed system, CalorieAi, provides users with a comprehensive platform to log food intake, track workout activities, monitor macronutrient distribution (proteins, carbohydrates, and fats), and receive intelligent insights based on their daily caloric balance. Through features such as real-time progress tracking, personalized goal setting, and adaptive AI-driven recommendations, the system CalorieAi aims to empower users to make informed, data-driven decisions about their health and fitness journey.

## **1.2 Background**

The integration of information technology into healthcare and fitness management has significantly transformed how individuals monitor and improve their physical well-being. Advances in digital technologies and increased internet accessibility have enabled the development of health applications that support informed lifestyle decisions. At the same time, sedentary lifestyles and unhealthy dietary habits have contributed to a rise in non-

communicable diseases such as obesity, diabetes, and cardiovascular conditions (World Health Organization, 2022).

A central concept in effective fitness management is energy balance, which refers to the relationship between caloric intake and energy expenditure. Maintaining an appropriate energy balance is essential for weight management and overall metabolic health. When caloric intake exceeds energy expenditure, weight gain occurs, while sustained caloric deficits lead to weight loss (Hall et al., 2012). Accurate estimation of energy expenditure and daily caloric needs is therefore critical for achieving **health** and fitness goals.

Energy expenditure varies across individuals due to factors such as age, sex, body composition, and physical activity levels. Traditional calorie estimation methods often rely on generalized equations that fail to reflect individual differences, leading to inaccurate feedback. Although digital fitness applications and wearable technologies have improved accessibility to health monitoring, many existing systems rely on simplified algorithms that limit personalization and accuracy (Krebs & Duncan, 2015).

Recent advancements in artificial intelligence (AI) offer opportunities to address these limitations. AI-powered systems can analyze user-specific data to generate personalized calorie estimates and adaptive recommendations. Studies indicate that AI-driven health applications like CalorieAi improve accuracy, transparency, and user engagement when compared to traditional systems (Chen & Zhang, 2020). Furthermore, web-based platforms enhance accessibility by enabling cross-device usage across different operating systems and device types without reliance on specialized hardware, thereby improving inclusivity and scalability of digital health systems (Fielding & Taylor, 2017; Topol, 2019).

Despite these advancements, challenges such as lack of transparency, limited adaptability, and data privacy concerns persist in existing solutions (Gao et al., 2015). In response, this study

focuses on the design and development of an AI-powered web application, CalorieAi, that integrates validated metabolic models with intelligent data processing to provide accurate, personalized, and transparent monitoring of energy expenditure and daily caloric needs.

### **1.3 Statement of the Problem**

Accurate estimation of energy expenditure and daily caloric requirements remains a significant challenge for individuals seeking to maintain, lose, or gain weight effectively. Energy expenditure is influenced by multiple factors, including basal metabolic rate, body composition, age, sex, and physical activity intensity, making precise estimation complex. However, many existing digital fitness and calorie tracking systems rely on generalized equations and population averages, which often fail to account for individual variability and lead to inaccurate guidance (Li & Mensah, 2023).

Inaccurate calorie estimations can negatively affect users' decision-making and health outcomes. Individuals attempting weight loss may unknowingly consume excess calories due to overestimated energy expenditure, while those aiming for weight gain may under-consume because of underestimated requirements. Such discrepancies can reduce user confidence, motivation, and long-term adherence to fitness goals (Li & Mensah, 2023).

Another limitation of existing systems is their narrow focus on exercise-related calorie burn without adequately integrating total daily energy expenditure. Components such as basal metabolism and non-exercise activities are often overlooked, leaving users without clear guidance on whether to maintain a calorie deficit, surplus, or balance (Ainsworth et al., 2011). Additionally, many applications operate as “black-box” systems, providing little explanation of how calorie estimates are generated, which undermines user trust and sustained engagement (Gao et al., 2015). Accessibility further constrains the effectiveness of current solutions, as



many systems depend on mobile-only platforms or wearable devices that may not be affordable or convenient for all users.

These challenges highlight the need for a transparent, accessible, and personalized system, such as CalorieAi, that integrates validated metabolic models with adaptive AI techniques. A web-based, AI-powered solution like CalorieAi offers a practical approach to delivering accurate, interpretable, and user-centered calorie monitoring, thereby supporting informed decision-making and sustainable health behavior change.

## **1.4 Study Objectives**

### **1.4.1 General Objective**

The primary objective of this project is to design, develop, and implement an intelligent web-based calorie tracking and energy monitoring system (CalorieAi) that provides users with accurate, transparent, and personalized tools to manage their dietary intake, monitor energy expenditure, and achieve their weight management goals through scientifically validated methods and explainable artificial intelligence.

### **1.4.2 Specific Objectives**

#### **IT-Related Objectives**

1. To design a responsive a web interface using HTML5, CSS3, and JavaScript (ES6+) that allows users to log workouts and track caloric data (ISO 9241-11).
2. To develop a comprehensive food logging system that tracks daily caloric intake across breakfast, lunch, and dinner with detailed macronutrient breakdown (proteins, carbohydrates, fats).
3. To implement a dynamic data loop that allows users to update body metrics (weight) in real-time, triggering automatic recalibration of metabolic calculations without session disruption.

4. To integrate a Generative AI computer vision model (Google Gemini API) that allows users to log dietary intake automatically by uploading images of their meals, dynamically extracting caloric and macronutrient estimations (Chen & Zhang, 2020).

### **Non-IT Objectives**

1. To help users make informed health and nutrition decisions through accurate, personalized caloric guidance (WHO, 2020).
2. To promote consistent exercise habits by providing clear performance feedback and motivational insights (Krebs & Duncan, 2015).

### **1.5 Scope of the Project**

This project focuses on the design and development of a web application, CalorieAi, that applies AI methods to monitor energy expenditure and estimate daily calorie needs (Chen & Zhang, 2020). It will define the data inputs, modeling approach, and evaluation criteria needed to support accurate estimates.

This study examines the development of a web-based system, CalorieAi, in which users enter personal information, including age, weight, height, gender, and level of physical activity, to generate estimates related to calorie needs and other related outputs (Lyons et al., 2014). The system estimates calories burned during physical activity by applying standard metabolic equivalent (MET) values (Ainsworth et al., 2011). It estimates daily energy requirements using validated metabolic equations, such as the Mifflin–St Jeor formula (Mifflin et al., 1990). The application provides recommendations on whether a user should maintain a calorie deficit, a calorie surplus, or calorie balance, based on the user’s weight management goals (Hall et al., 2012).

The system uses artificial intelligence methods to support personalization by updating its recommendations based on user input and observed patterns of activity (Esteva et al., 2019).

The project includes secure user authentication, reliable data storage, and basic data visualization that allows users to monitor progress over time.

This study focuses on a web-based system, CalorieAi, developed using HTML5, CSS3, and JavaScript (ES6+), with Firebase used for the backend infrastructure (Fielding & Taylor, 2017). The system was tested and assessed with a group of students from the University of Professional Studies, Accra (UPSA), focusing on usability, accuracy, and system performance.

The scope does not include integration with wearable fitness devices, development of a mobile application, training of advanced machine learning models, or deployment of the system at large commercial scale. These constraints keep the project manageable within the academic schedule and still allow the development of a working solution that meets the research aims.

## **1.6. Methodology for Project**

This study uses a design and development method to build an AI-based web application, CalorieAi, and to assess its performance during actual use in real-world settings (Richey & Klein, 2007). The development process starts by collecting and documenting the project requirements. At this stage, the team identifies functional and non-functional requirements by reviewing relevant research and assessing user expectations. These requirements shape system design decisions and remain important throughout implementation.

The system, CalorieAi, is built on common web technologies. HTML5 defines the page structure, CSS3 controls the layout and visual style, and JavaScript (ES6+) implements the application logic and user interaction (Fielding & Taylor, 2017). In this project, Firebase serves as the backend, providing user authentication, data storage, and real-time data processing.

The system applies a mixed approach in artificial intelligence that joins rule-based reasoning with formulas validated in peer-reviewed studies to provide clear estimates of calorie needs and related dietary guidance (Chen & Zhang, 2020; Mifflin et al., 1990). The project uses an Agile method, progressing through short iterations that include design work, testing, and revisions to the system (Beck et al., 2001). This approach uses feedback from each testing phase to guide continued revisions to the system.

The system, CalorieAi, was evaluated using questionnaire responses, observation notes from the test sessions, and data captured in the system logs (Lewis, 1995). This study applies descriptive statistics and qualitative analysis to assess usability, accuracy, and user satisfaction. The method evaluates the system by examining technical performance and user acceptance, then combines the results into a single summary score.

### **1.7 Significance of the Project**

The value of this study can be considered in terms of technological, practical, academic, and societal aspects. From a technical standpoint, the study combines artificial intelligence with current web technologies to improve the performance and usability of digital health applications (Fielding & Taylor, 2017). By pairing AI-based reasoning with metabolic equations that have been validated in prior research, the proposed system, CalorieAi, goes beyond fixed, rule-based calorie estimates and instead provides decision support that can adjust to individual data over time (Esteva et al., 2019). This framing supports the practical role of hybrid AI methods in health systems (Chen & Zhang, 2020).

In practical terms, the study offers people clear and usable information about their energy use and their daily calorie needs through CalorieAi. By stating whether a person is in a calorie deficit, surplus, or balance, the system helps users make informed choices about weight loss,

weight maintenance, or weight gain. This is consistent with the results reported by Hall et al. In 2012, they argue that successful weight management requires a clear understanding of energy intake and energy expenditure, along with consistent management of their balance (Hall et al., 2012). Because the system runs on the web, users can access it from different devices and do not need specialized hardware.

This study adds to the research literature on digital health, fitness monitoring, and AI-driven healthcare applications. It sets out a clear framework through CalorieAi that brings together metabolic science, user-centered design, and intelligent computation, and it can guide future research and system development. This study contributes to current debates on transparency, usability, and user trust in AI-based health systems (Gao et al., 2015).

From a societal standpoint, this study supports preventive health care by increasing awareness of calorie needs and encouraging healthy lifestyle choices. As lifestyle-related diseases increase worldwide, there is a growing need for digital health tools like CalorieAi that are accessible and reliable to support better public health outcomes (WHO, 2022).

In sum, this study adds to research on intelligent, inclusive, and user-centered digital health tools.

### **1.8 Limitations of the Project**

While CalorieAi, the proposed AI-powered web application provides a practical way to track energy expenditure and estimate daily caloric needs, it also has limitations that should be stated clearly. A central limitation is that the system depends on user-entered data, including age, weight, height, and information about activity levels. Calorie estimates produced by the system may be less accurate when users enter incorrect or incomplete data. The application does not include wearable sensors or real-time physiological measures, such as heart rate. Calorie

expenditure is calculated using standard metabolic equations and MET values (Mifflin et al., 1990; Ainsworth et al., 2011). Although these methods are supported by research, they may not fully reflect differences in individual metabolism or changes in exercise intensity.

While the system incorporates advanced Generative AI (Large Language Models and computer vision) for automated food image recognition, the accuracy of image-based macronutrient estimation depends heavily on photo quality, lighting, and the visual clarity of mixed meals (Esteva et al., 2019). Furthermore, this generative AI feature requires an active internet connection to communicate with external APIs. Lastly, while the system effectively tracks live workout sessions, it relies on browser local storage rather than backend server caching for active timer persistence, meaning cross-device resumption of a live workout is not currently supported (Sommerville, 2016). Consequently, while the system ensures local persistence, active workout timers do not synchronize across multiple devices in real-time.

User testing was limited to a small group, mainly students at the University of Professional Studies, Accra, which may limit how well the findings apply to other populations (Lewis, 1995). Because it is web-based, the application requires stable internet access to run as intended. Network disruptions can reduce access and may prevent some users from using real-time features as intended.

## **1.9 Organization of the Study**

This study is organized into five chapters. Each chapter addresses one part of the study, so the research is presented in a clear order. Chapter One introduces the study and sets out its overall purpose and scope. This section provides the research background, states the problem, and sets

out the study objectives and research questions. It explains why the study matters, clarifies the project scope and limits, and ends by describing how the report is structured.

Chapter Two provides a review of related literature. This chapter reviews the main theories, research findings, and existing systems on monitoring energy expenditure, estimating calories, applying artificial intelligence in health settings, and understanding how users adopt digital health technologies (Lyons et al., 2014; Chen & Zhang, 2020). It also outlines the theoretical framework that guides the study.

Chapter Three outlines the research methods used in this study. It describes the research design and approach, the study population and sampling methods, the data collection methods, and the steps used to analyze the data (Richey & Klein, 2007). This chapter describes the design, development, and evaluation of the system, CalorieAi.

Chapter Four describes the system implementation of CalorieAi and presents the testing process and evaluation results. The text outlines the system architecture, the main implementation strategies, and the testing methods used (Sommerville, 2016). It also describes the issues that arose during development and the steps taken to address them so the system works as intended.

Chapter Five summarizes the study's main findings, states the conclusions based on the results, notes the study's limitations, and offers recommendations for future research. This chapter explains the project's contribution to information technology and digital health solutions.

## **1.10 Chapter Summary**

This chapter outlined the study that designed and developed an AI-based web application, CalorieAi, to track energy expenditure and estimate daily calorie needs. The study opened by

noting that digital health technologies are becoming more relevant for helping people manage their fitness and nutrition goals with greater success (Krebs & Duncan, 2015).

This chapter examined the main limitations of current calorie tracking systems, focusing on problems with estimation accuracy, limited personalization, and unclear methods for calculating calorie values (Li & Mensah, 2023). The study stated its objectives to describe the purpose and intended outcomes of the research, and it set research questions that guided the design and assessment of the proposed system, CalorieAi.

The study explained why this work matters by describing its role in information technology through the use of artificial intelligence and web-based methods (Chen & Zhang, 2020). It also noted the practical value for users who need trustworthy support for managing health and fitness (Lyons et al., 2014).

The project's scope and limitations were defined to set clear boundaries for the study. The chapter ended by describing the report's structure, setting out an organized plan for the chapters that follow. Chapter One established the basis for the study and clarified the focus for the literature review and research methods presented in the chapters that follow.



## **CHAPTER TWO**

### **LITERATURE REVIEW**

#### **2.1 Introduction**

This chapter surveys the scholarly literature that informs the design and development of an AI-based web application, CalorieAi, that tracks energy expenditure and estimates daily calorie requirements. The review places the study in the wider research area of digital health and fitness management (Krebs & Duncan, 2015). This section reviews the main ideas behind estimating calorie intake, understanding energy balance, and using artificial intelligence in systems designed for health contexts (Chen & Zhang, 2020). This study reviews empirical research on current calorie tracking and fitness monitoring tools to evaluate how well they work and where they fall short (Li & Mensah, 2023). The chapter examines theories of technology adoption and user engagement in digital health applications (Lyons et al., 2014). This chapter reviews prior studies to identify gaps in existing systems, with a focus on accuracy, personalization, and transparency. These gaps support the need for the proposed solution, CalorieAi. This review provides an academic basis for the system design choices and the research method used in the study.

#### **2.2 Theoretical Review**

The design and development of the CalorieAi system is grounded in several established theoretical frameworks drawn from technology adoption, health behavior, and human–computer interaction research. These theories provide the academic rationale for the system's core design decisions and inform the evaluation criteria used to measure its effectiveness. The Technology Acceptance Model (TAM), proposed by Davis (1989), is one of the most widely applied models for understanding user adoption of information systems. TAM posits that two primary factors determine whether users will accept and use a new technology: Perceived

Usefulness (PU) and Perceived Ease of Use (PEOU). Perceived Usefulness refers to the degree to which a user believes that using the system will enhance their performance in a given task, while Perceived Ease of Use refers to the degree to which a user believes that the system will be free from effort (Davis, 1989).

In the context of CalorieAi, Perceived Usefulness is addressed through the system's ability to deliver accurate, personalized calorie estimates using validated metabolic models such as the Mifflin–St Jeor equation (Mifflin et al., 1990), as well as its capacity to provide actionable guidance on caloric deficit, surplus, or maintenance in alignment with individual fitness goals (Hall et al., 2012). Perceived Ease of Use is addressed through the integration of the Google Gemini Vision API for automated food scanning, which significantly reduces the manual effort traditionally associated with calorie logging, and through the implementation of an intuitive, responsive glassmorphic user interface designed in accordance with established usability principles (ISO 9241-11, 2018; Fielding & Taylor, 2017). TAM provides a strong theoretical foundation for evaluating whether users find CalorieAi both useful and easy to use, which are critical determinants of sustained engagement with digital health systems (Krebs & Duncan, 2015).

The Health Belief Model (HBM), originally developed by Rosenstock (1974) and later expanded by Janz and Becker (1984), is a psychological framework that explains and predicts health-related behaviors based on individual beliefs about health threats and the perceived benefits of taking preventive action. The model identifies several key constructs, including perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cues to action, and self-efficacy (Rosenstock, 1974; Janz & Becker, 1984). In the context of this study, the HBM informs the system's design by recognizing that users are more likely to adopt and consistently use CalorieAi if they perceive a genuine health risk associated with poor dietary

habits or sedentary lifestyles, such as the rising prevalence of obesity and non-communicable diseases (World Health Organization, 2022). The system addresses perceived benefits by providing transparent, real-time feedback on energy balance that enables users to see the direct consequences of their dietary and exercise choices. Perceived barriers, such as the inconvenience and time required for manual food logging, are mitigated through the AI-powered plate scanning feature that automates macronutrient estimation (Esteva et al., 2019; Chen & Zhang, 2020). Cues to action are embedded within the system through daily AI-generated insights, progress notifications, and gamification elements that prompt users to engage with the application regularly (Lyons et al., 2014). Self-efficacy is reinforced through achievable goal milestones, visual progress charts, and dynamic badge achievements that build user confidence in their ability to manage their energy balance effectively over time (Bandura, 1986).

Social Cognitive Theory (SCT), proposed by Bandura (1986), emphasizes the role of observational learning, self-regulation, and self-efficacy in shaping human behavior. SCT posits that behavior is influenced by the continuous interaction between personal cognitive factors, behavioral patterns, and environmental influences, a concept referred to as reciprocal determinism (Bandura, 1986). In the context of CalorieAi, SCT informs the design of the gamification engine and the predictive analytics module. The system's achievement badges function as behavioral reinforcement mechanisms that reward users for consistent engagement, such as logging meals daily or completing weekly workout targets, thereby strengthening positive health behaviors through self-regulatory feedback loops (Lyons et al., 2014). The predictive weight-change analytics, which forecast weekly weight outcomes based on the established principle that a net energy balance of approximately 7,700 kilocalories equates to one kilogram of body mass (Hall et al., 2012), enable users to observe the projected consequences of their current behaviors before those consequences materialize. This aligns

with SCT's emphasis on outcome expectations as a driver of behavioral change (Bandura, 1986). Furthermore, the system's transparent presentation of BMR and TDEE calculations serves an educational function that enhances self-efficacy by helping users understand the scientific basis of their caloric needs, rather than presenting results as opaque or arbitrary values (Chen & Zhang, 2020).

The Theory of Planned Behavior (TPB), developed by Ajzen (1991), extends the Theory of Reasoned Action by incorporating the construct of perceived behavioral control alongside attitudes and subjective norms as predictors of behavioral intention. TPB suggests that an individual's intention to perform a health-related behavior, such as maintaining a calorie deficit for weight loss, is influenced by their attitude toward the behavior, the social norms surrounding the behavior, and their perceived ability to successfully perform the behavior (Ajzen, 1991). CalorieAi applies TPB principles by shaping positive user attitudes through clear, data-driven feedback that demonstrates the tangible benefits of calorie monitoring. The system reinforces perceived behavioral control by providing users with comprehensive tracking tools and actionable recommendations that simplify the process of managing energy balance, thereby reducing the perceived complexity of dietary management (Hall et al., 2012; Krebs & Duncan, 2015). Together, these theoretical frameworks provide a comprehensive academic foundation for the design, development, and evaluation of CalorieAi.

### **2.3 Review of Research Variables**

The evaluation of the CalorieAi system requires the identification and definition of key research variables that influence the system's effectiveness and user outcomes. These variables are categorized as independent variables, dependent variables, and intervening variables, consistent with standard research methodology practices (Creswell & Creswell, 2018). The independent variables in this study represent the factors that are manipulated, controlled, or

introduced through the system's design and functionality. The first independent variable is the accuracy of the metabolic calculation models employed by the system, specifically the Mifflin–St Jeor equation for estimating Basal Metabolic Rate and the standardized Metabolic Equivalent of Task values for calculating exercise-related energy expenditure (Mifflin et al., 1990; Ainsworth et al., 2011). The second independent variable is the integration of the Google Gemini Vision API for automated food recognition and macronutrient estimation, which represents the AI-driven component of the system's calorie logging functionality (Esteva et al., 2019; Chen & Zhang, 2020). The third independent variable is the implementation of gamification elements, including achievement badges and predictive weight-change forecasting, which serve as behavioral reinforcement mechanisms designed to sustain user engagement over time (Lyons et al., 2014; Hall et al., 2012). The fourth independent variable is the quality and responsiveness of the user interface, which directly affects user interaction patterns and task completion efficiency (ISO 9241-11, 2018; Nielsen, 2012).

The dependent variables represent the outcomes or effects that the study seeks to measure and attribute to the influence of the independent variables. The first dependent variable is the accuracy of calorie estimation, defined as the degree to which the system's computed caloric values align with outputs derived from manually applied validated metabolic formulas (Mifflin et al., 1990; Westerterp, 2013). The second dependent variable is user engagement, measured by the frequency and consistency of user interactions with core system features such as food logging, workout recording, and dashboard analytics review (Krebs & Duncan, 2015). The third dependent variable is user satisfaction, assessed through structured questionnaires and usability evaluation instruments that capture user perceptions of the system's usefulness, ease of use, and relevance to personal fitness goals (Davis, 1989; ISO 9241-11, 2018). The fourth dependent variable is the level of nutritional awareness demonstrated by users, defined as the

user's ability to identify macronutrient composition and make informed dietary decisions after sustained interaction with the system (Hall et al., 2012).

Several intervening variables may influence the relationship between the independent and dependent variables and must therefore be acknowledged. These include the user's prior knowledge of nutrition and fitness concepts, which may affect their interpretation of system outputs regardless of the system's accuracy (Westerterp, 2013). The reliability and speed of the user's internet connection may influence system responsiveness, particularly for features that depend on real-time communication with external APIs such as Google Gemini or Firebase Cloud Firestore (Fielding & Taylor, 2017). The accuracy of self-reported biometric data, including body weight, height, and exercise intensity, also functions as an intervening variable, since calculation accuracy depends fundamentally on the quality of user-provided inputs (Prince et al., 2008; Li & Mensah, 2023). Additionally, external environmental factors, such as the availability of healthy food options and the user's daily schedule constraints, may affect the extent to which system recommendations translate into observable behavioral changes. Acknowledging these intervening variables is essential for interpreting evaluation outcomes accurately and for identifying potential confounding factors that may limit the generalizability of the study's findings (Creswell & Creswell, 2018).

## **2.4 Contextual Background of the Study Area**

The study is situated within the University of Professional Studies, Accra (UPSA), a public university in Ghana that specializes in professional and career-oriented tertiary education. UPSA serves a student population of approximately 20,000 individuals alongside academic and administrative staff (UPSA, 2023). The university is located in the Greater Accra Region, the most urbanized region in Ghana, and its student body is drawn from diverse socioeconomic, cultural, and geographical backgrounds across the country. The contextual relevance of UPSA

as the study area is informed by several converging factors related to public health, digital technology adoption, and the specific lifestyle challenges faced by tertiary-level students.

Ghana, like many other Sub-Saharan African countries, is experiencing a significant epidemiological transition characterized by the increasing prevalence of diet-related non-communicable diseases (NCDs) such as obesity, type 2 diabetes, and cardiovascular conditions (World Health Organization, 2022). This transition has been driven in part by rapid urbanization, changes in dietary patterns toward processed and energy-dense foods, and declining levels of physical activity among young adults (Bosu, 2015). Data from the Ghana Health Service indicate that the prevalence of overweight and obesity among urban-dwelling adults has risen significantly over the past two decades, with university-aged populations identified as particularly vulnerable due to the adoption of sedentary academic lifestyles, irregular eating patterns, and limited access to nutritional guidance (Ofori-Asenso et al., 2016).

Within the university context, students often face time constraints that discourage structured meal planning and regular physical activity. Academic workloads, combined with the availability of convenient but calorie-dense foods from campus vendors and nearby commercial outlets, contribute to poor dietary habits that may persist beyond the university years (Krebs & Duncan, 2015). Furthermore, many students lack formal education in nutrition science and may not be equipped to accurately estimate their caloric intake or energy expenditure without the assistance of supportive tools (Li & Mensah, 2023). These factors underscore the relevance of developing a digital health solution, CalorieAi, tailored to the needs and constraints of university students operating within an urban Ghanaian context.

From a technological perspective, UPSA provides a suitable environment for evaluating a web-based health application due to the high level of digital device ownership and internet usage among its student population. Smartphone penetration among Ghanaian university students has

increased substantially in recent years, and most students regularly access web-based platforms for academic, social, and transactional purposes (Agyemang et al., 2020). The university campus is equipped with institutional internet infrastructure, and students commonly utilize mobile data connections to access online services. This level of digital readiness supports the viability of deploying and testing CalorieAi as a web-based solution that does not require specialized hardware or proprietary mobile applications, consistent with the system's design principle of hardware-independent accessibility (Fielding & Taylor, 2017). By setting the study within UPSA, the research addresses a real-world health challenge affecting a defined and accessible population, while simultaneously leveraging the digital readiness of the study area to ensure meaningful system evaluation (Bosu, 2015).

## **2.5 Review of Existing Systems and Technologies**

Several digital fitness and calorie tracking systems have been developed to help individuals monitor physical activity, estimate energy expenditure, and manage their weight. These systems are commonly implemented as mobile applications, wearable-device platforms, or web-based solutions. Although they have contributed significantly to health awareness and self-monitoring, existing systems exhibit notable limitations that motivate the development of the proposed AI-powered web application.



| <b>System</b> | <b>Platform Type</b>                   | <b>Key Features</b>                                                                       | <b>Strengths</b>                                                | <b>Limitaions Identified</b>                                                                                                                                                              | <b>Relevance to Proposed System</b>                                                     |
|---------------|----------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| MyFitnessPal  | Mobile & Web Application               | Food logging, physical activity tracking, calorie balance estimation, large food database | Easy to use, extensive food database, basic progress tracking   | Relies on generalized metabolic equations; limited personalization for activity intensity and individual metabolism; low transparency in calorie calculation logic (Krebs & Duncan, 2015) | Highlights need for personalized calorie estimation and transparent computational logic |
| Google Fit    | Mobile Application & Wearable Platform | Automatic activity tracking, step counting, GPS-based activity                            | Sensor-based data collection; integration with multiple devices | Uses simplified activity models leading to variable accuracy; limited support                                                                                                             | Informs need for integrated energy expenditure and daily caloric needs analysis         |

|                                      |                                                  |                                                                               |                                                                                              |                                                                                                                                    |                                                                                       |
|--------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                                      |                                                  | monitoring,<br>calorie<br>estimation                                          |                                                                                              | for estimating<br>daily caloric<br>needs for<br>weight<br>management<br>(Li & Mensah,<br>2023)                                     |                                                                                       |
| Apple<br>Health/<br>Apple<br>Fitness | Mobile<br>Application &<br>Wearable<br>Ecosystem | Activity<br>tracking,<br>calorie<br>estimation,<br>health data<br>aggregation | High data<br>accuracy due<br>to wearable<br>sensors;<br>seamless<br>ecosystem<br>integration | Requires<br>proprietary<br>hardware;<br>limited<br>accessibility;<br>opaque calorie<br>estimation logic<br>(Lyons et al.,<br>2014) | Emphasizes need for<br>hardware-independent<br>and accessible web-<br>based solutions |

## 2.6 Proposed System

CalorieAi is a web application that uses AI to track users' energy expenditure and estimate their daily calorie needs, with attention to personalization, transparency, and ease of use. This system was designed to address limits in current fitness and calorie-tracking applications, including reliance on broad estimates, limited adjustment to individual users, unclear methods, and barriers to access (Li & Mensah, 2023). In contrast to many existing systems that depend only on fixed equations or hard-to-interpret algorithms, this system, CalorieAi, combines

scientifically validated metabolic models with hybrid artificial intelligence logic (Chen & Zhang, 2020).

Energy expenditure during physical activity is often estimated by using standard Metabolic Equivalent of Task (MET) values, which offer a consistent way to estimate calories burned across different activities (Ainsworth et al., 2011). Daily calorie needs are estimated with the Mifflin–St Jeor equation, which is commonly used to approximate resting energy expenditure (Mifflin et al., 1990). These models provide the core reasoning of the system and support its scientific validity.

The system's artificial intelligence adjusts calorie recommendations to the individual by updating them in response to user inputs such as age, gender, height, weight, activity type, duration, and fitness goals (Chen & Zhang, 2020). A hybrid AI approach that combines rule-based logic with deterministic formulas can keep the decision process transparent while still producing recommendations that fit the given context. To further address the friction of manual data entry, CalorieAi integrates a Generative AI scanning feature. By leveraging Google's Gemini API, the system utilizes computer vision to analyze user-uploaded images of meals, automatically extracting food names, calories, and macronutrient breakdowns (Esteva et al., 2019).

Additionally, to improve user retention, the system employs gamification through a dynamic achievements engine that awards badges based on real-time activity metrics (Lyons et al., 2014). Predictive analytics are also implemented to forecast weekly weight changes, operating on the established physiological principle that a net energy balance of approximately 7,700 kcal equates to one kilogram of body mass (Hall et al., 2012). This points to a central limitation

in current platforms: users often do not know how these systems generate calorie estimates (Chen & Zhang, 2020; Gao et al., 2015).

The system, CalorieAi, is implemented as a web-based application and can be accessed through standard web browsers on desktop computers, tablets, and smartphones. This approach avoids reliance on proprietary mobile platforms or wearable devices, which may improve access and support inclusion (Fielding & Taylor, 2017). The front end is built with HTML5, CSS3, and JavaScript (ES6+) to support a clear user interface with responsive interactions. Firebase provides the back-end services, including authentication, data storage, and real-time updates (Fielding & Taylor, 2017).

Along with estimating calories burned, CalorieAi includes an analysis of daily caloric needs. This helps users judge whether a calorie deficit, surplus, or balance fits their fitness goals (Hall et al., 2012). The feedback and recommendations are personalized and shown in clear visual formats to support informed decisions and sustained engagement over time (Lyons et al., 2014). By correcting limits in existing systems and prioritizing accuracy, transparency, and accessibility, CalorieAi provides a clearer and more user-focused approach to tracking energy expenditure and calorie intake.

## **This chapter 2.7 Chapter Summary**

summarized prior research and outlined the methods used to design an AI-based web application, CalorieAi, that records energy expenditure and estimates daily calorie requirements. This study reviewed core ideas in digital health and fitness management and found that an increasing number of people rely on digital tools to track calorie intake (Krebs & Duncan, 2015). Previous systems were examined and compared using the features each study reported, along with the limitations described in earlier research (Li & Mensah, 2023).

The review identified repeated issues: the options offered did not align with individual users' needs, the calorie estimates lacked specificity, and the method used to produce the reported results was not explained in a clear and transparent way (Gao et al., 2015). The review identified gaps in these systems that reduce user trust and make it harder to maintain engagement over time.

To address these limits, we developed a web-based system, CalorieAi, that applies AI to improve the accuracy of calorie tracking, support monitoring at the individual level, and extend access to a wider group of users. This chapter set out the foundation of the study and informed the methodological and design choices discussed in the following chapter.

## **CHAPTER THREE**

### **METHODOLOGY**

#### **3.1 Introduction**

This chapter describes the methods used to design and build an AI-based web application, CalorieAi, that tracks energy expenditure and estimates daily calorie needs. This section outlines the study's research design and overall approach, and it describes the target population, the sampling methods used, and the procedures for collecting data (Creswell & Creswell, 2018).

This chapter explains the methods used to analyze the data and the system modeling techniques applied during the development process (Richey & Klein, 2007). This chapter lays out the study's methods in a clear order so that the work follows the stated aims and can be reviewed and repeated.

#### **3.2 Research Design**

This study employs a mixed-methods research design that integrates both quantitative and qualitative data collection and analysis techniques within a Design and Development Research (DDR) framework. Mixed-methods research combines the strengths of quantitative measurement with the depth of qualitative inquiry, enabling a comprehensive evaluation of both system performance and user experience (Creswell & Creswell, 2018). This approach is particularly appropriate for studies involving the development and evaluation of technology-based interventions, where technical functionality must be assessed alongside user perceptions, usability, and behavioral outcomes (Richey & Klein, 2007).

The Design and Development Research framework, as defined by Richey and Klein (2007), is a systematic approach to the study of design, development, and evaluation processes with the aim of establishing an empirical basis for the creation of products and tools. DDR emphasizes

iterative cycles of planning, construction, and evaluation, making it fundamentally compatible with the Agile development methodology adopted for this project (Beck et al., 2001). Within this framework, the quantitative component of the research design involves the collection of numerical data through structured questionnaires that measure user satisfaction, perceived usefulness, perceived ease of use, and system accuracy on predefined scales (Davis, 1989; ISO 9241-11, 2018). These data are analyzed using descriptive statistical techniques to identify patterns and assess whether the system meets its specified performance benchmarks (Creswell & Creswell, 2018).

The qualitative component involves the collection of observational data during usability testing sessions, including field notes on user navigation behavior, verbal feedback provided during task completion, and identification of interface friction points. These qualitative insights complement the quantitative measurements by providing contextual explanations for observed patterns and informing iterative system refinements (Rubin & Chisnell, 2008; Sharp, Rogers, & Preece, 2019). The integration of quantitative and qualitative methods through a convergent parallel design allows the research to achieve triangulation, whereby findings from one data source are verified and enriched by findings from another source, thereby strengthening the overall validity and reliability of the study's conclusions (Creswell & Creswell, 2018). This research design is consistent with established practices in health informatics and software engineering research, where multi-method evaluation strategies are recommended to capture both the measurable and experiential dimensions of system quality (Pressman & Maxim, 2020; Sommerville, 2016).

### **3.3 Population**

The target population for this study comprises the academic community of the University of Professional Studies, Accra (UPSA), which includes undergraduate and postgraduate students

as well as academic and administrative staff. The total university population is approximately 20,000 students and over 1,000 staff members, drawn from diverse academic faculties including Information Technology, Business Administration, Accounting, and Law (UPSA, 2023).

This population is considered appropriate for the study for several reasons. First, university students and staff represent a demographic group that is increasingly affected by lifestyle-related health challenges, including sedentary behaviors and poor dietary habits, which are directly relevant to the objectives of the CalorieAi system (World Health Organization, 2022; Bosu, 2015). Second, the population exhibits a high level of digital literacy and regular engagement with web-based platforms, which supports the feasibility of evaluating a browser-based health application that requires no specialized hardware or wearable devices (Fielding & Taylor, 2017).

Third, the diversity of the university population, which includes individuals of varying ages, genders, body compositions, activity levels, and fitness goals, provides a heterogeneous sample that enables the assessment of system performance across a range of user profiles, consistent with the study's emphasis on personalized calorie estimation and adaptive recommendations (Hall et al., 2012; Westerterp, 2013). The accessible population, which constitutes the subset of the target population that can be practically reached for participation in the study, includes students and staff who are present on the main UPSA campus during the study period and who have access to web-capable devices such as smartphones, laptops, or tablets (Creswell & Creswell, 2018). This accessible population provides a sufficient and representative base from which the study sample is drawn.



### **3.5 Sample and Sampling Techniques**

A sample of thirty (30) participants was selected from the accessible population at the University of Professional Studies, Accra, using a purposive sampling technique. Purposive sampling, also known as judgmental sampling, is a non-probability sampling method in which participants are deliberately selected based on their relevance to the research objectives and their ability to provide meaningful data for the study (Creswell & Creswell, 2018; Etikan et al., 2016). This technique was deemed appropriate for the study because the evaluation of CalorieAi requires participants who possess specific characteristics that align with the system's intended user base and who are capable of providing informed feedback on system functionality, usability, and relevance.

The inclusion criteria for participant selection required that individuals met the following conditions: (1) current enrollment as a student or current employment as a staff member at the University of Professional Studies, Accra; (2) ownership of or regular access to a web-capable device with a functioning camera, as the AI plate scanning feature requires image capture capability; (3) self-identified interest in monitoring dietary intake, physical activity, or body weight management, ensuring relevance of the system's features to the participant's personal goals; and (4) willingness to engage with the CalorieAi system during the evaluation period and to provide honest feedback through questionnaires and observation sessions (Creswell & Creswell, 2018).

Exclusion criteria included individuals with no interest in fitness or dietary tracking, individuals without access to a web-capable device, and individuals unable to participate during the designated evaluation period. The sample of thirty participants was distributed to ensure diversity across several demographic dimensions, including gender, age range, academic faculty, and self-reported fitness goal (weight loss, weight maintenance, or weight gain). This distribution was intended to enable the assessment of system performance and

usability across a representative range of user profiles, consistent with the study's emphasis on personalization and adaptability (Davis, 1989; Hall et al., 2012). While purposive sampling does not support statistical generalization to the broader population, it is widely accepted in exploratory and developmental research where the objective is to generate in-depth insights about system performance within a defined user group (Etikan et al., 2016; Creswell & Creswell, 2018). The findings derived from this sample are intended to inform system refinement and to establish a foundation for larger-scale evaluations in future research.

### **3.6 Data Collection Instruments**

Data collection for this study was conducted using three primary instruments: structured questionnaires, the System Usability Scale, and direct observation with checklists. The use of multiple data collection instruments supports methodological triangulation, which enhances the validity and reliability of the study's findings by enabling cross-verification of results obtained from different sources (Creswell & Creswell, 2018).

**Structured Questionnaires:** A pre-evaluation questionnaire was administered to all participants prior to their initial interaction with the CalorieAi system. This instrument collected demographic information (age, gender, academic faculty), baseline health awareness data (current dietary habits, exercise frequency, prior experience with calorie tracking applications), and self-reported fitness goals (weight loss, weight maintenance, or weight gain). Following the evaluation period, a post-evaluation questionnaire was administered to capture participants' perceptions of the system's usefulness, accuracy, relevance, and overall impact on their understanding of energy balance. The post-evaluation questionnaire employed a five-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5) to measure key constructs aligned with the Technology Acceptance Model, including perceived usefulness and perceived ease of use (Davis, 1989). The questionnaire also included open-ended questions that allowed participants to provide qualitative comments on specific system

features, including the AI plate scanning module, the gamification badges, and the predictive weight-change analytics of CalorieAi. Both questionnaires were administered via Google Forms to facilitate efficient distribution and data compilation (Creswell & Creswell, 2018).

**System Usability Scale (SUS):** The System Usability Scale, developed by Brooke (1996), is a standardized ten-item questionnaire that provides a reliable and quick assessment of the perceived usability of a system. The SUS employs a five-point Likert scale for each item and produces a composite score ranging from 0 to 100, where a score above 68 is generally considered above average and indicative of acceptable usability (Brooke, 1996; Bangor et al., 2009). The SUS was selected because it is technology-agnostic, has been validated across a wide range of digital health applications, and provides a quantitative benchmark that can be compared against published norms for web-based systems (Bangor et al., 2009; ISO 9241-11, 2018). The SUS was administered to all participants after they completed a predefined set of core tasks within the CalorieAi application, including registering and completing the onboarding process, logging a meal using both manual entry and AI plate scanning, starting and completing a workout session using the interactive timer, reviewing dashboard analytics and AI-generated insights, and updating body weight and reviewing predictive weight-change forecasts.

**Direct Observation with Checklists:** During the pilot evaluation sessions, members of the research team conducted structured observations of participants as they interacted with the CalorieAi system. An observation checklist was used to record quantitative metrics such as time-on-task for each core function, the number of errors or missteps encountered during task completion, and instances where participants required external guidance or clarification. Qualitative observations were also documented, including navigation patterns, points of hesitation or confusion, and spontaneous verbal feedback offered by participants during interaction. This observational approach is consistent with established usability evaluation

methods in human–computer interaction research, where direct observation provides rich contextual data that cannot be captured through self-report instruments alone (Rubin & Chisnell, 2008; Nielsen, 2012; Sharp, Rogers, & Preece, 2019). The combination of quantitative observation metrics and qualitative field notes enabled the research team to identify specific interface elements and interaction flows that required refinement, thereby directly informing the iterative system improvement process aligned with the Agile development methodology adopted for this project (Beck et al., 2001; Sommerville, 2016).

### **3.7 System Development Methodology**

The AI-based web application, CalorieAi, was developed using the Agile system development methodology (Beck et al., 2001; Sommerville, 2016). Agile was selected because it supports iterative and incremental development, allowing project requirements and system features to evolve as better understanding is gained throughout the development process (Pressman & Maxim, 2020). This characteristic aligns strongly with a user-centered approach to system design, where continuous user feedback informs design decisions and system refinement. Unlike linear development methodologies such as the Waterfall model, Agile allows changes to requirements and functionalities to be incorporated even after development has begun, without significantly disrupting the project workflow (Sommerville, 2016).

This flexibility makes Agile particularly suitable for projects in which user needs are expected to change over time and where system quality is improved through repeated cycles of testing, evaluation, and refinement (Pressman & Maxim, 2020). Agile is a project management and system development approach that divides development activities into short, repeated cycles known as sprints, with each sprint delivering a functional component of the system (Beck et al., 2001). These iterative cycles promote early detection of design issues, improve

collaboration between stakeholders, and enhance the overall quality and usability of the developed system (Schwaber & Sutherland, 2020).

In each cycle, the team delivers a small, usable output, so the project moves forward through a sequence of incremental releases over time. In each sprint, the team delivers a testable increment of the system, such as a user login feature, activity logging, calorie estimation, or a feedback screen. This approach helps identify design flaws, usability issues, and missing functions at an early stage, so changes can be made before the project moves into later phases of development. This study views Agile as a useful approach because it allows developers to collect user feedback at planned intervals and adjust the work while development is ongoing.

Because CalorieAi focuses on individualized support, straightforward use, and accurate calorie tracking, we gathered user feedback to refine both the core logic and the interface design. Agile methods allowed testing and review in each iteration, which helped keep new features consistent with user expectations and the project's goals. In agile development, teams often work on system models, coding, testing, and documentation in parallel. This overlap may shorten development time and reduce the project's overall risk (Pressman & Maxim, 2020).

The methods were designed to remain flexible so that validated metabolic models and hybrid AI rules could be added or adjusted as project requirements changed over the course of development. In contrast, approaches such as the Waterfall model were often considered less suitable because their fixed sequence of phases leaves limited room to respond to change. Prototyping alone was insufficient because it did not provide a clear plan for developing the full system or a clear approach for evaluating its performance. Agile was chosen for the CalorieAi web application because it supports short development cycles, makes it easier to revise work as requirements change, and keeps user needs central in design and implementation decisions.

### **3.8 Crystallization of the Problem**

The crystallization of the problem involved a systematic analysis of challenges associated with existing fitness and calorie monitoring systems, with the aim of clearly defining the core issues the proposed system seeks to address. Initial analysis revealed that many current calorie tracking applications rely heavily on generalized metabolic equations and static assumptions, which fail to adequately capture individual differences in age, body composition, activity intensity, and fitness goals (Ainsworth et al., 2011; Westerterp, 2013; Li & Mensah, 2023). Studies have shown that such generalized approaches often result in inaccurate calorie estimations, reducing the effectiveness of these systems for weight management and fitness planning (Westerterp, 2013; Li & Mensah, 2023).

Another key issue identified was the limited level of personalization in existing systems. Many applications provide uniform recommendations that do not dynamically adjust to changes in user behavior or evolving fitness objectives. Users seeking to lose, gain, or maintain weight are often not provided with clear guidance on caloric deficits or surpluses, making it difficult to translate system outputs into actionable decisions (Krebs & Duncan, 2015). This gap highlighted the need for a system capable of generating personalized caloric recommendations based on user-specific inputs and defined goals.

Transparency in calorie estimation logic was also identified as a significant concern. Several widely used fitness platforms operate as black-box systems, offering little explanation of how calorie values are generated. Research indicates that lack of transparency in health-related systems reduces user trust and negatively affects long-term engagement (Lyons et al., 2014; Gao et al., 2015). Addressing this challenge required the design of a system, CalorieAi, that utilizes scientifically validated formulas and clearly communicates the basis of its calculations to users.

Accessibility further emerged as a critical issue during the problem analysis. Many existing fitness platforms depend on proprietary mobile applications or wearable devices, which limit access for users without specialized hardware. This constraint informed the decision to adopt a web-based solution, as web applications offer broader accessibility across devices and platforms without additional hardware requirements (Fielding & Taylor, 2017). These identified problems directly influenced the design and development of the proposed CalorieAi web application. By integrating validated metabolic equations with hybrid AI logic, the system improves accuracy while maintaining transparency (Ainsworth et al., 2011; Chen & Zhang, 2020). The web-based architecture enhances accessibility, and the inclusion of goal-oriented feedback addresses user decision-making needs. Through this process of problem crystallization, the study translated observed shortcomings in existing systems into concrete system requirements and design features that underpin the proposed solution.

### **3.9 Requirements of the Proposed System**

The proposed system, CalorieAi, is expected to provide a secure and user-friendly web platform for monitoring energy expenditure and daily caloric needs, consistent with best practices in digital health system design (ISO/IEC 25010, 2011). It should allow users to register, log in, and input personal and fitness-related information using secure authentication mechanisms to protect sensitive health data (OWASP, 2021). The system should enable users to record physical activities and log meals accurately in order to calculate calories burned using validated metabolic models, which are widely recognized as reliable approaches for estimating energy expenditure (Ainsworth et al., 2011; Hall et al., 2012).

In addition, the system must estimate daily caloric requirements and provide guidance on caloric surplus, deficit, or maintenance in alignment with individual fitness goals, as recommended in energy balance and weight management research (Hall et al., 2012).

Personalized feedback and progress visualization should be generated to support informed decision-making and sustained user engagement, which are critical factors in effective digital health interventions (Krebs & Duncan, 2015). The system, CalorieAi, must also be accessible across multiple devices through standard web browsers, thereby improving usability and inclusivity for diverse user groups (Fielding & Taylor, 2017).

Furthermore, data security and user privacy must be ensured through secure data storage, encryption, and access control mechanisms, as emphasized in health information security guidelines (ISO/IEC 27001, 2013). The application should deliver reliable performance and timely system responses to enhance user satisfaction and trust (Pressman & Maxim, 2020). Overall, the system must be scalable, maintainable, and capable of future enhancements in order to support long-term use and system evolution (Sommerville, 2016).

### **3.9.1 Functional Requirements**

The proposed AI-powered web application, CalorieAi, is designed to provide comprehensive functionality that supports secure user access, personalized onboarding, accurate calorie monitoring, intelligent feedback, and reliable data management. The functional requirements of the system are described as follows:

**User Authentication:** The system shall provide secure user authentication mechanisms to ensure controlled access to the application, in line with established web security best practices (OWASP, 2021). It shall allow new users to register using a valid email address and password and securely authenticate existing users during login using encrypted credentials (Stallings, 2017). The system shall also support password recovery functionality to enable users to reset forgotten credentials securely. To enhance usability and continuity, the system shall maintain active user sessions across browser sessions while enforcing appropriate session management and security controls (ISO/IEC 27001, 2013). Furthermore, email and password authentication



is securely managed by Firebase, which handles all password hashing. Each user session is actively monitored using Firebase's `onAuthStateChanged()` listener. If an unauthorized user attempts to access the dashboard, this event-driven architecture automatically intercepts the request and redirects them to the login page (Pressman & Maxim, 2020).

**User Onboarding:** Upon successful registration, the system shall guide users through an onboarding process to collect essential biometric and lifestyle information, including age, gender, height, and body weight, which are required for accurate metabolic calculations (Hall et al., 2012). The system shall allow users to select fitness goals such as weight loss, weight maintenance, or weight gain and assess their typical activity level. Based on the collected data, the system shall automatically calculate the user's Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE) to support personalized calorie recommendations using validated metabolic equations (Mifflin et al., 1990; Hall et al., 2012).

**Food Logging:** The system, CalorieAi, shall enable users to log daily meals, including breakfast, lunch, and dinner, in order to monitor dietary intake effectively. It shall track total caloric intake and macronutrient distribution, specifically proteins, carbohydrates, and fats, which are essential for nutritional assessment and weight management (Westerterp, 2013). A searchable food database shall be provided to facilitate accurate and efficient meal logging, while custom food entry functionality shall allow users to record foods not available in the predefined database (Krebs & Duncan, 2015).

**Exercise Tracking:** The system shall support physical activity tracking by allowing users to log various forms of exercise. It shall support at least twenty different activity types and calculate calories burned using standardized Metabolic Equivalent of Task (MET) values,

which are widely used for estimating energy expenditure during physical activity (Ainsworth et al., 2011). Users shall be able to record exercise duration and perceived intensity, and the system shall store workout history to support performance review and progress monitoring over time (Li & Mensah, 2023).

**Dashboard Analytics:** The system shall present an interactive dashboard that provides users with real-time analytics related to energy balance. Key metrics such as BMR, TDEE, total caloric intake, calories burned, and net calorie balance shall be displayed to support informed decision-making (Hall et al., 2012). Weekly activity and nutrition trends shall be visualized using charts to improve data interpretation and user engagement. The system shall also calculate and display goal progress percentages to help users monitor advancement toward defined fitness objectives (Krebs & Duncan, 2015).

**AI-Driven Insights:** The system, CalorieAi, shall incorporate artificial intelligence logic to generate personalized daily recommendations based on user behavior, historical data, and caloric balance. It shall provide alerts when users deviate significantly from their defined targets and suggest appropriate behavioral or lifestyle adjustments. AI-driven personalization has been shown to improve engagement and relevance in digital health systems by adapting recommendations to individual user patterns (Chen & Zhang, 2020).

**Data Management:** The system shall store all user data securely in a cloud-based database to ensure availability, scalability, and fault tolerance (Sommerville, 2016). User data shall be synchronized in real time to enable seamless access across sessions and devices. The system shall provide users with the option to delete their data in compliance with data protection and privacy principles (ISO/IEC 27001, 2013). Data integrity shall be maintained by ensuring

accuracy, consistency, and protection against unauthorized access or modification (Pressman & Maxim, 2020).

**AI Plate Scanning:** The system shall allow users to upload images of their meals. It shall utilize a third-party Generative AI API (Google Gemini) to analyze the image, identify food components, and generate JSON data containing caloric and macronutrient estimates (Esteva et al., 2019; Chen & Zhang, 2020).

**Live Session Tracking and Persistence:** The system shall provide an interactive stopwatch and countdown timer for physical activities. To prevent data loss during accidental browser refreshes, the active session state shall be persistently cached in the browser's local storage mechanism (Sommerville, 2016).

**Gamification and Predictive Analytics:** The system, CalorieAi, shall dynamically evaluate the user's weekly data loop to unlock achievement badges (Lyons et al., 2014). It shall also calculate and display forecasted weekly weight changes by dividing the user's net caloric balance by 7,700 kcal (Hall et al., 2012).

### **3.9.2 Non-Functional Requirements**

In addition to functional capabilities, the proposed AI-powered web application, CalorieAi, is required to meet several non-functional requirements that ensure optimal performance, usability, security, reliability, scalability, and maintainability. These requirements define the quality attributes of the system and are critical to its effectiveness, user satisfaction, and long-term sustainability.

**Performance:** The system shall be designed to deliver efficient and responsive performance under normal operating conditions, as recommended for interactive web applications (Nielsen, 2012). Page load times shall not exceed two seconds when accessed using standard broadband internet connections, since response delays beyond this threshold negatively affect user experience and engagement (Nielsen, 2012; Google Web Dev, 2020). The rendering of charts and graphical analytics shall occur within 500 milliseconds to ensure smooth user interaction and real-time feedback (Few, 2013). Database queries shall be optimized to complete within one second in order to support real-time data retrieval and processing in data-driven applications (Elmasri & Navathe, 2016).

**Usability:** The system shall provide an intuitive and user-friendly interface that can be easily understood by non-technical users, in accordance with established usability and human-computer interaction principles (ISO 9241-11, 2018). The application, CalorieAi, shall be fully responsive, enabling seamless access and interaction across mobile phones, tablets, and desktop devices, which is a core requirement for modern web-based systems (Marcotte, 2014). Error messages shall be clearly worded and actionable to guide users in resolving issues without confusion or technical complexity (Nielsen, 2012).

**Security:** The system shall implement strong security mechanisms to protect user data and prevent unauthorized access. User passwords shall be securely hashed and encrypted prior to storage using accepted cryptographic practices (Stallings, 2017). All data transmitted between the client and server shall be protected using HTTPS encryption protocols to safeguard data confidentiality and integrity (OWASP, 2021). Authentication tokens shall be configured to expire after a maximum duration of twenty-four hours to minimize the risk of session hijacking (OWASP, 2021). In addition to HTTPS communication, the system leverages user-based

Firestore security rules to restrict access to personal data. By enforcing UID-based document access control, the system ensures that a user can only query, read, or modify their own specific health records, thereby preventing unauthorized data exposure and ensuring strict confidentiality (Bass, Clements, & Kazman, 2012).

**Reliability:** The system shall be designed to operate reliably with minimal downtime. A target system uptime of 99.5 percent shall be maintained through the use of Firebase's cloud infrastructure, which provides built-in redundancy and fault tolerance (Google Cloud, 2023). User data shall be backed up automatically to prevent data loss, and the system shall implement graceful error-handling mechanisms that notify users appropriately in the event of system failures or service interruptions (Sommerville, 2016).

**Scalability:** The system, CalorieAi, shall be capable of supporting increased usage without degradation in performance. It shall support at least one thousand concurrent users while maintaining acceptable response times, which is a common scalability benchmark for web-based information systems (Pressman & Maxim, 2020). Database operations shall be optimized using indexing and query optimization techniques to ensure efficient data access as the user base grows (Elmasri & Navathe, 2016).

**Maintainability:** The system shall be developed using clean, well-structured, and maintainable code to support long-term system evolution. Consistent naming conventions shall be applied throughout the codebase to enhance readability and reduce maintenance effort (Sommerville, 2016). Functions and modules shall be designed to be reusable and loosely coupled, enabling easier updates and future enhancements (Pressman & Maxim, 2020).

Comprehensive technical documentation shall be maintained to support system maintenance, debugging, and future development activities (IEEE, 2018).

### 3.9.3 Software Requirements

| Component               | Description                                                                                                                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend Technologies   | HTML5, CSS3, and JavaScript (ES6+) are required to design and implement an interactive, responsive, and user-friendly web interface. (Mozilla Developer Network , 2023; World Wide Web Consortium , 2022). |
| Backend Services        | Firebase is required to provide secure user authentication, real-time database management, and cloud-based data storage for user profiles and activity records. (Google Firebase, 2023).                   |
| Development Environment | A modern code editor such as Visual Studio Code is required for writing, testing, debugging, and managing the application source code efficiently. (Microsoft, 2023).                                      |
| Web Browser             | Modern web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge are required for testing, running, and                                                                                       |

|                       |                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | validating system functionality across platforms. (Mozilla Developer Network, 2023).                                                                                                              |
| Version Control Tools | Git and GitHub are required for source code version management, collaboration among developers, and tracking changes throughout the development lifecycle. (Chacon & Straub, 2014; GitHub, 2023). |
| External APIs         | Google Gemini Flash API is required to process image-to-text generative AI requests for the automated food scanning module.                                                                       |
| Data Visualization    | Chart.js is required to render responsive, interactive bar and line charts for tracking weekly caloric trends and historical weight progression.                                                  |
| UI/UX Libraries       | Animate On Scroll (AOS) and FontAwesome are required to provide smooth interface transitions and scalable vector icons.                                                                           |
|                       |                                                                                                                                                                                                   |

### 3.9.4 Hardware Requirements

| Component                  | Minimum Specification                                                                 | Recommended Specification                                                                       |
|----------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Processor                  | Intel Core i3 or AMD Ryzen 3(Intel, 2023; AMD, 2023)                                  | Intel Core i5 or AMD Ryzen 5(Intel, 2023; AMD, 2023)                                            |
| Random Access Memory (RAM) | 4GB(Mozilla Developer Network, 2023)                                                  | 8 GB or higher(Mozilla Developer Network, 2023)                                                 |
| Storage                    | 256 GB Hard Disk Drive (HDD) (Silberschatz et al., 2018)                              | 512 GB Solid State Drive (SSD) (Silberschatz et al., 2018)                                      |
| Display                    | Screen resolution of 1366 × 768 pixels(W3C, 2022)                                     | Screen resolution of 1920 × 1080 pixels or higher(W3C, 2022)                                    |
| Network                    | Broadband internet connection with a minimum speed of 1 Mbps(Fielding & Taylor, 2017) | Broadband internet connection with a minimum speed of 5 Mbps or higher(Fielding & Taylor, 2017) |
|                            |                                                                                       |                                                                                                 |

### 3.10 Design of the System

The proposed AI-powered web application, CalorieAi, was designed through a structured system analysis process that focused on identifying user needs and translating those needs into clear, testable system specifications. The work started by identifying the main problems people face when tracking energy expenditure accurately and estimating daily calorie needs, based on a review of prior studies, an assessment of existing systems, and reported user expectations (Li & Mensah, 2023). The main issues identified were limited personalization, low transparency in how calorie values were calculated, and weak integration between activity tracking and daily calorie targets (Lyons et al., 2014; Gao et al., 2015). From this analysis,



functional and non-functional requirements were established to ensure the system is accurate, easy to use, secure, and accessible (ISO/IEC 25010, 2011). User requirements, including activity logging, calorie calculation, goal-based recommendations, and progress tracking, were translated into corresponding system functions which directly shaped the final design of CalorieAi.

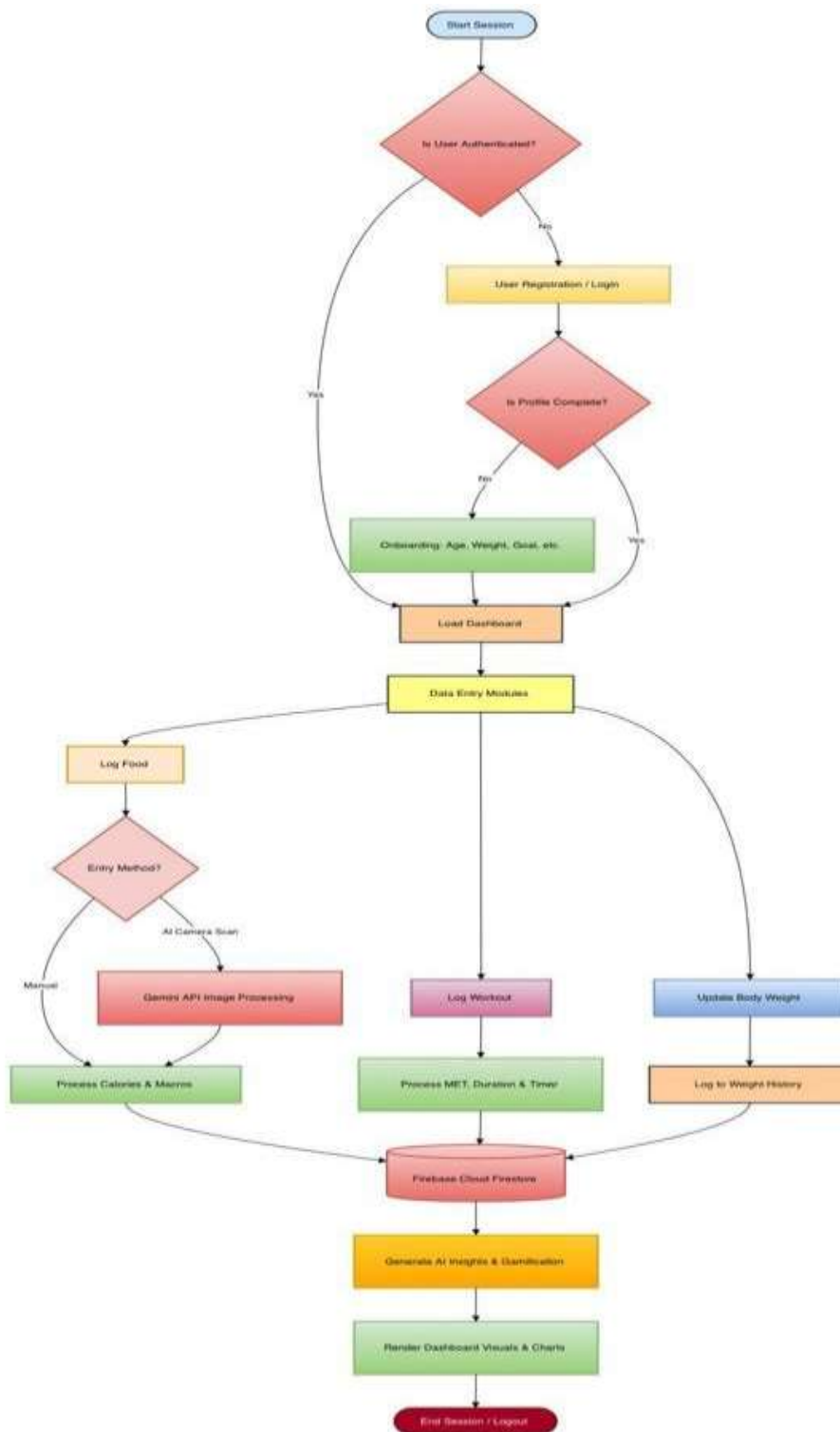
CalorieAi uses a web-based setup that collects user input and calculates calorie values using scientifically validated mathematical models. Specifically, the system calculates the Basal Metabolic Rate (BMR) using the Mifflin-St Jeor Equation:

$BMR = (10 \times W) + (6.25 \times H) - (5 \times A) + S$ , where  $W$  is weight in kg,  $H$  is height in cm,  $A$  is age in years, and  $S$  is +5 for biological males or -161 for biological females (Mifflin et al., 1990). To determine Total Daily Energy Expenditure (TDEE), this BMR is multiplied by an activity factor (1.2 for sedentary, 1.375 for lightly active, 1.55 for moderately active, and 1.725 for very active) (Westterp, 2013). The results are then stored in an organized database to ensure consistency and reliability in user tracking.

The design of CalorieAi uses a modular structure in which the user interface, application logic, and data storage are separated into distinct components that work together through clearly defined interfaces. This approach supports long-term maintenance, future growth, and clear system operation (Sommerville, 2016). The system was described in both logical and visual terms using several modeling tools, including flowcharts, context diagrams, entity relationship diagrams (ERD), data flow diagrams (DFD), and use case diagrams (Pressman & Maxim, 2020). Together, these diagrams outline how users work with the system, how data moves through its parts, and how information is stored and handled to meet the research objectives.

### **3.10.1 Flowchart Diagram**

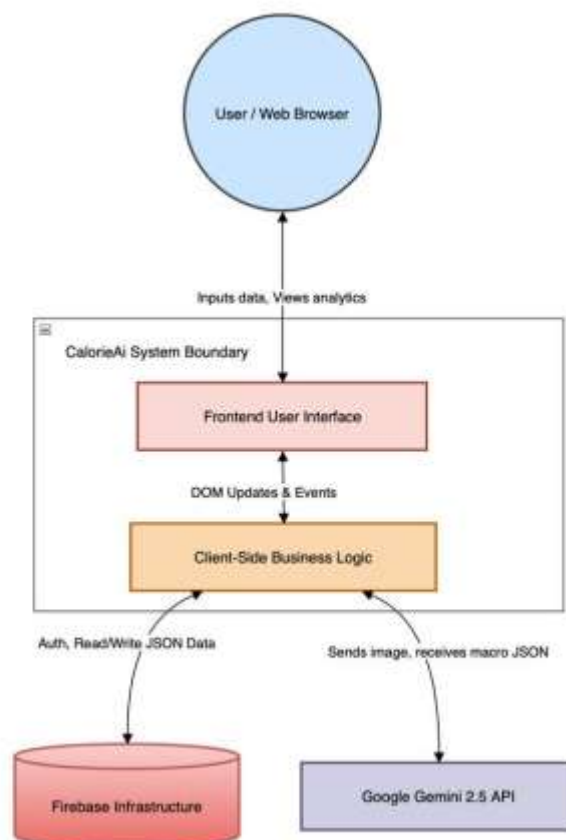
This section outlines the system's logical flow from initial input to final output. The process begins when the user accesses the web application and encounters the authentication gateway. If unauthenticated, the user registers or logs in. New users are routed through a mandatory onboarding sequence to capture baseline biometrics (age, weight, height, gender, and activity level) required for metabolic calculations. Once authenticated and profiled, the user accesses the main dashboard, which serves as the hub for data entry. The user can choose to log food (either manually or via the Gemini AI camera scanner), log a workout using the interactive session timer, or update their body weight. These inputs are processed locally via Javascript logic to calculate caloric values and macronutrients, and are then securely transmitted to Firebase Cloud Firestore. Finally, the system's analytics engine retrieves this updated data to generate predictive AI insights, unlock gamification badges, and render visual progress charts before the user ends their session.



**Figure 1: System Flow Chart Diagram (Thompson & Sykes, 2026).**

### 3.10.2 Context Diagram

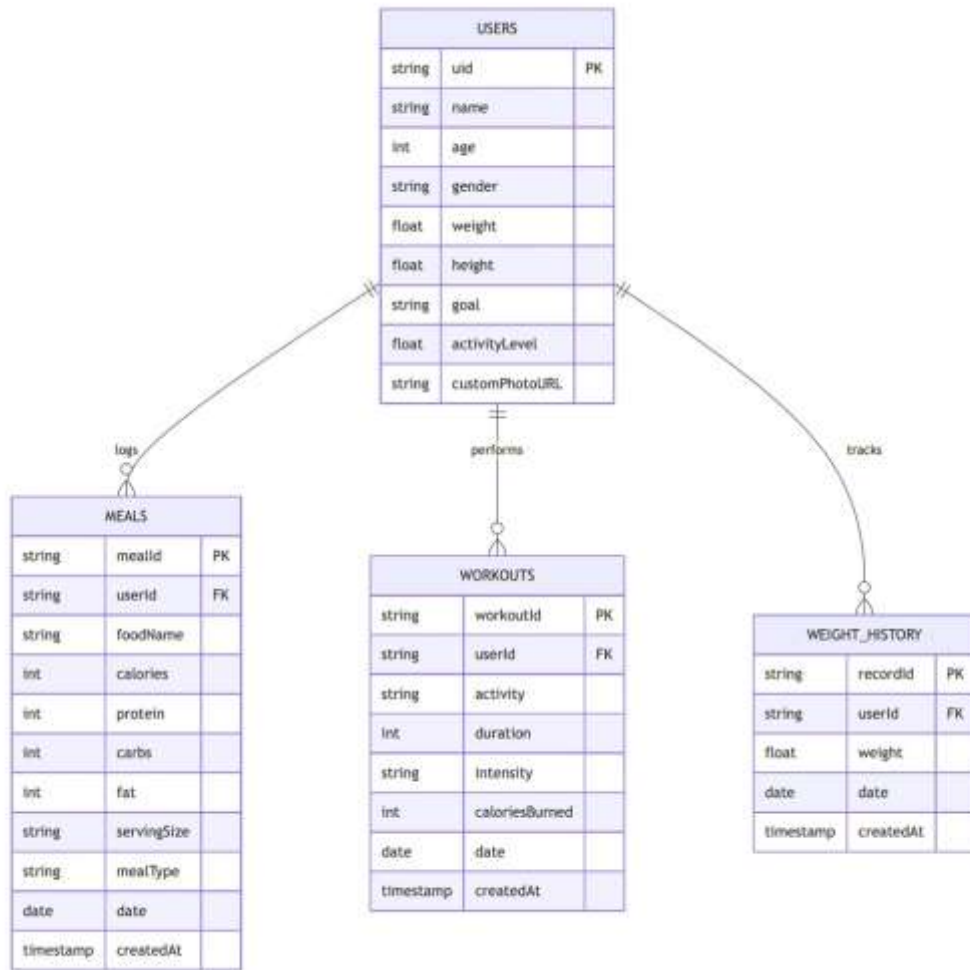
The CalorieAi context diagram illustrates the system boundaries, major external actors, and the data exchanged between them. The application utilizes a serverless, client-cloud architecture model (Fielding & Taylor, 2017). Users access the system exclusively through web browsers, where the frontend UI and all client-side business logic operate. Rather than relying on a traditional middle-tier server, the browser communicates directly via secure API calls with two external dependencies: Firebase and the Google Gemini API. Firebase acts as the backend infrastructure, providing secure authentication and NoSQL database management. Concurrently, the application sends base64-encoded meal images to the external Google Gemini API, which processes the images via computer vision and returns formatted JSON data containing macronutrient estimations.



**Figure 2 : Context Diagram (Thompson & Sykes, 2026).**

### **3.10.3 Entity Relationship Diagram (ERD)**

The CalorieAi system utilizes Firebase Firestore, a NoSQL cloud-hosted database, which structures data using collections and documents to ensure system scalability (Sommerville, 2016). The database schema is built around four main entities. The Userscollection acts as the primary record, storing profile information, goal metrics, and biometric data uniquely identified by the Firebase Authentication UID (Primary Key). The Meals collection records daily caloric intake, macronutrient distribution, and serving sizes. The Workouts collection logs physical activity, intensity multipliers, and active timer durations. Finally, a WeightHistory collection is utilized to track longitudinal weight fluctuations over time. A strict one-to-many relationship enforces data integrity, allowing a single user to record multiple meals, workouts, and weight updates while keeping all documents linked securely to their unique user ID (Foreign Key).

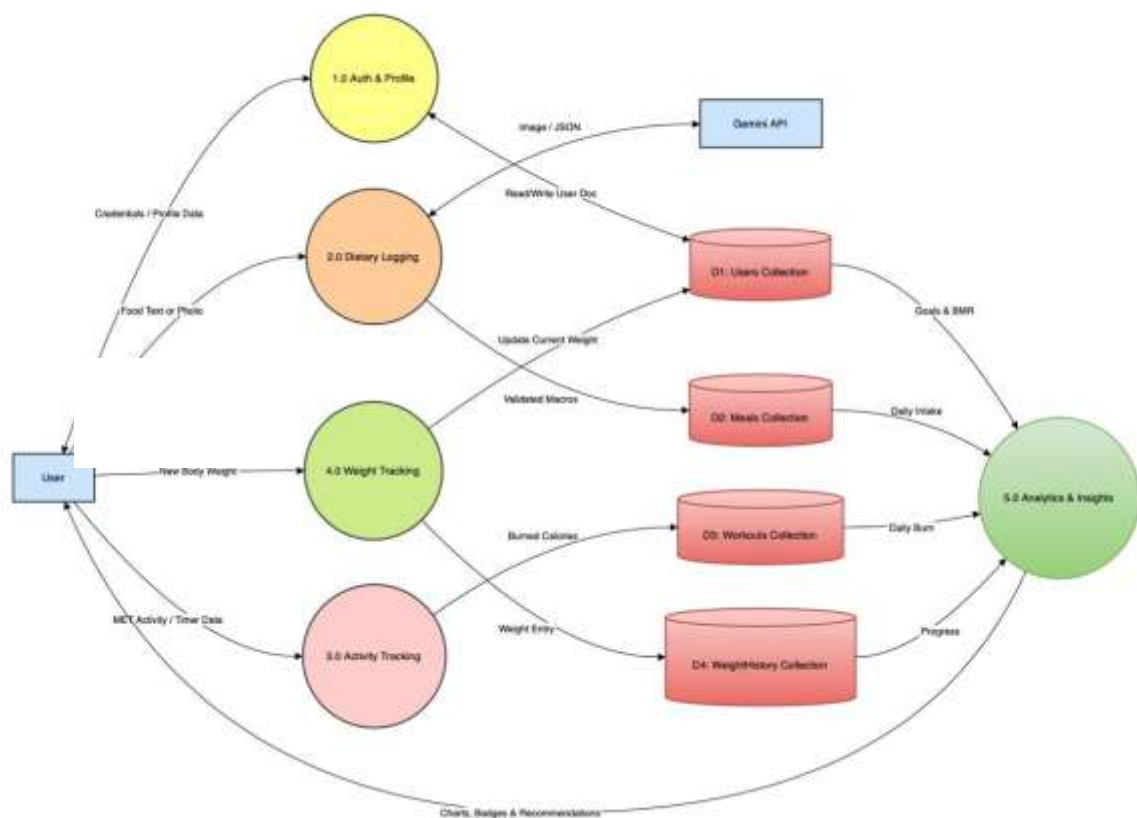


**Figure 3: ERD Diagram (Thompson & Sykes, 2026).**

### 3.10.4 Data Flow Diagram (DFD)

The CalorieAi Data Flow Diagram (DFD) maps the movement of information between the user, the core processing modules, and the external data stores. Users input raw data via credentials, text, live timer sessions, or meal photographs. The Dietary Logging process handles food inputs; if a photo is uploaded, it is routed to the Gemini API for visual processing before being validated and stored in the Meals Collection. The Activity Tracking process calculates energy expenditure by applying MET values to the user's timer data, pushing the output to the Workouts Collection. The Weight

Tracking process pushes new biometric entries to the WeightHistory Collection while simultaneously updating the user's baseline in the Users Collection. Finally, the Analytics & Insights process continuously pulls real-time data from all four Firestore collections to compute net energy balances, evaluate gamification thresholds, and render predictive visual charts back to the user interface.

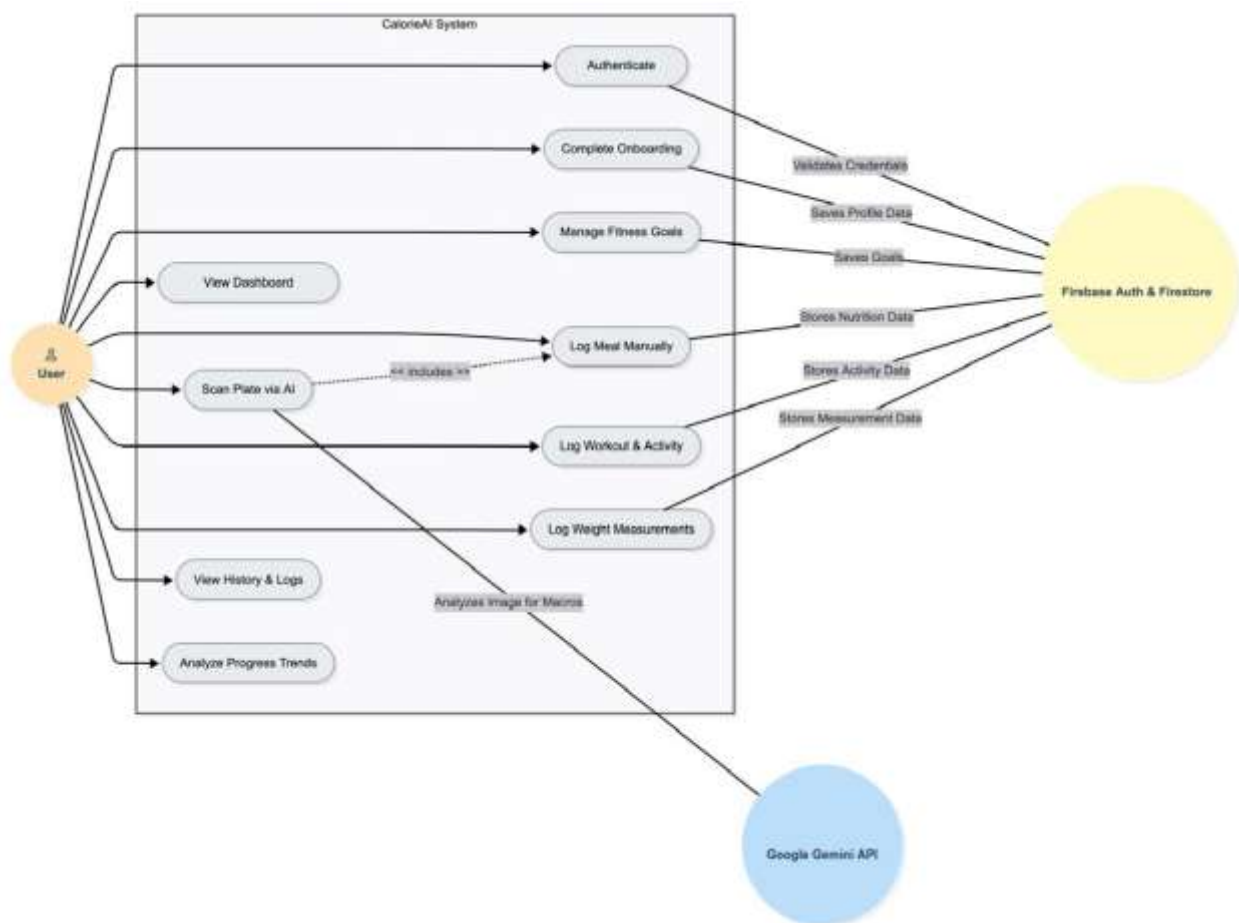


**Figure 4: Data Flow Diagram (Thompson & Sykes, 2026).**

### 3.10.5 Use Case Diagram

The CalorieAi use case diagram maps the core functional requirements and system interactions, all centered around a single primary actor: the User. The diagram defines the functional scope of the project, highlighting nine primary actions the user can execute within the application boundary. The authentication and setup services include Register/Login and Complete Onboarding. The primary data entry functions encompass

Logging Food (with sub-options for manual entry or AI camera scanning), Logging Workouts (utilizing integrated session timers), and Updating Body Weight. The visualization and feedback functions allow the user to View Dashboard Charts and View AI Insights & Badges. Finally, the user retains administrative control through the Edit Profile Settings use case. Each feature operates synergistically to support the ultimate goal of data-driven health management.



**Figure 5: Use Case Diagram (Thompson & Sykes, 2026).**



### **3.11 Chapter Summary**

This chapter described the design of the proposed AI-powered web application for monitoring energy expenditure and daily caloric needs. It explained the system analysis process used to identify user requirements and translate them into system specifications. The conceptual design of the system was presented, emphasizing modularity, usability, and scalability. Various system modeling tools, including flowcharts, context diagrams, ERD, DFD, and use case diagrams, were discussed to illustrate system structure, data flow, and user interaction. Overall, the system design provides a clear blueprint that guides the development and implementation of the proposed solution.

## **CHAPTER FOUR**

### **IMPLEMENTATION AND DOCUMENTATION OF THE PROPOSED SYSTEM**

#### **4.1 Introduction**

Delivering a software system requires more than design; it also requires careful implementation, systematic testing, and clear documentation to ensure quality and reliability (Sommerville, 2016). This chapter describes the implementation of an AI-based web application, CalorieAi, designed to track energy expenditure and estimate daily calorie requirements. It explains how the system design and requirements were transformed into a functional web application through the selection and application of appropriate web and cloud technologies (Pressman & Maxim, 2020).

The implementation process involved integrating the user interface, backend services, and AI-based calorie calculation logic into a single cohesive system, which is consistent with best practices in web-based system development (Sommerville, 2016). Rigorous testing was carried out to verify the system's functionality, accuracy, usability, and performance, as systematic testing is essential for identifying defects and ensuring compliance with system requirements (IEEE, 2018). Multiple testing techniques were applied to detect and correct errors prior to deployment, thereby improving system stability and reliability (Pressman & Maxim, 2020).

User interaction testing further indicated that CalorieAi was easy to use and aligned with user expectations, supporting established principles of usability and user-centered design (ISO 9241-11, 2018). Comprehensive documentation was produced to describe system features, implementation procedures, and testing outcomes, serving as a reference for system maintenance, scalability, and future enhancements (IEEE, 2018). In summary, this chapter

demonstrates that the developed system is reliable, usable, and suitable for practical real-world application.

## **4.2 Results (Non-IT Related)**

This section presents the non-IT-related results of the study, focusing on the health behavioral outcomes, nutritional awareness gains, and fitness engagement patterns observed among participants during the evaluation of the CalorieAi system. While the technical performance of the system is addressed separately through IT-specific testing approaches, the non-IT results provide insight into the broader impact of the application on user health awareness and behavior (Krebs & Duncan, 2015).

Analysis of the post-evaluation questionnaire data revealed that the majority of participants reported a measurable increase in their awareness of daily caloric intake and energy expenditure patterns after using the CalorieAi system. Prior to interacting with the system, most participants indicated limited knowledge of key nutritional concepts such as Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE). Following sustained use of the application, participants demonstrated an improved ability to estimate the caloric content of familiar foods and to articulate the relationship between energy intake and energy expenditure (Hall et al., 2012; Westerterp, 2013).

The AI-powered plate scanning feature, which utilizes the Google Gemini Vision API, was identified by participants as particularly impactful in increasing dietary awareness. Several participants reported that the automated food analysis revealed previously unrecognized sources of caloric intake, particularly from beverages and condiments. This observation aligns with research demonstrating that computer vision-based dietary assessment methods can identify nutritional components that are commonly overlooked (Esteva et al., 2019; Chen & Zhang, 2020).

Regarding physical activity engagement, participants who consistently interacted with the workout logging module and the interactive session timer reported increased motivation (Lyons et al., 2014). The system's real-time calculation of calories burned, derived from standardized MET values (Ainsworth et al., 2011), provided participants with immediate quantitative feedback. The gamification elements, including the dynamic achievement badges, were reported to function as effective motivational reinforcements (Bandura, 1986; Lyons et al., 2014). Furthermore, the predictive weight-change analytics module helped shape dietary decisions by providing a forward-looking perspective (Hall et al., 2012; Chen & Zhang, 2020).

#### **4.3 Discussion (Non-IT Related)**

The non-IT-related results provide evidence that CalorieAi is effective not only as a technical system but also as a behavioral health intervention. The observed improvement in nutritional awareness supports the Health Belief Model's proposition that providing individuals with clear information about health risks and benefits can motivate positive action (Rosenstock, 1974; Janz & Becker, 1984). By presenting transparent calculations of BMR and TDEE, the system transforms abstract concepts into concrete, personally relevant data points (Lyons et al., 2014; Chen & Zhang, 2020).

The finding that participants identified previously unrecognized caloric sources through AI plate scanning highlights the educational value of computer vision-based assessment. By automating food identification, CalorieAi reduces reliance on self-reporting, which is often subject to bias (Prince et al., 2008; Esteva et al., 2019). The positive impact of gamification is consistent with prior research indicating that behavior change techniques like progress feedback are effective in promoting sustained interaction (Lyons et al., 2014). These dynamic badges function as operant reinforcers within the framework of Social Cognitive Theory (Bandura, 1986).

The influence of predictive weight-change forecasting aligns with the Theory of Planned Behavior, which holds that behavioral intention is strengthened when individuals perceive a clear connection between current actions and future outcomes (Ajzen, 1991). From a public health perspective, CalorieAi has the potential to function as a preventive health tool, especially given the rising rates of obesity in urban Ghana (Bosu, 2015; World Health Organization, 2022). Its web-based architecture ensures accessibility without specialized hardware, lowering the barrier to entry (Fielding & Taylor, 2017).

#### **4.4 Testing Approaches**

Software testing is a core component of the software development lifecycle, particularly for applications like CalorieAi that handle sensitive health-related data and generate decision-support outputs, where reliability and accuracy are critical (Sommerville, 2016; Pressman & Maxim, 2020). In this study, the AI-based web application was systematically tested to evaluate its accuracy, reliability, usability, and overall readiness for deployment. Since the system tracks energy expenditure and estimates daily caloric requirements, inaccurate or inconsistent outputs could reduce user trust and negatively influence health-related decision-making (Westerterp, 2013).

To address these risks, a structured, multi-layer testing strategy was adopted. The testing process evaluated both the technical performance of the system and its usability from the end-user perspective. It verified that individual system components functioned correctly, that system operations conformed to the specified functional requirements, and that users could interact with the application effectively without unnecessary difficulty (IEEE, 2018). Additionally, the testing process assessed whether the final system was suitable for practical, real-world use.

This study employed four testing approaches: unit testing, functional testing, usability testing, and acceptance testing. Each testing method focused on a specific quality attribute of the system, and collectively they formed a comprehensive evaluation framework consistent with established software engineering best practices (Pressman & Maxim, 2020).

#### **4.4.1 Unit Testing**

Unit testing involved the systematic examination of individual components of the AI-powered web application, CalorieAi, to verify that each module performed its intended function correctly when isolated from the rest of the system (Sommerville, 2016). This testing approach was particularly important for validating the accuracy of the system's computational logic, which forms the core of calorie estimation and daily caloric need calculations (Pressman & Maxim, 2020).

Key system components tested at the unit level included the user authentication module, profile data processing functions, workout logging mechanisms, energy expenditure calculation logic, and daily caloric requirement computation. Each unit was tested using predefined input values and expected outputs to ensure consistency and correctness, in line with standard unit testing practices (IEEE, 2018). For example, the calorie calculation module was tested by supplying known MET values, workout durations, and user characteristics, and comparing the output against manually computed results derived from established metabolic models such as the Mifflin–St. Jeor equation and standardized MET tables (Mifflin et al., 1990; Ainsworth et al., 2011).

This approach ensured that the system's outputs aligned with scientifically validated formulas used in exercise physiology and nutrition science. Unit testing also helped identify logical errors, incorrect conditional statements, and data validation issues early in the development process (Pressman & Maxim, 2020). By detecting faults at the component level, corrective

measures were implemented before integration with other system modules, thereby reducing the likelihood of cascading errors during later testing phases. Additionally, unit testing contributed to improved code quality and maintainability. Ensuring that each module functioned independently and correctly allows future enhancements or modifications to be implemented with reduced risk of unintended side effects (Sommerville, 2016). Overall, unit testing served as the foundation upon which reliable and robust system functionality was built.

#### **4.4.2 Functional Testing**

Functional testing was carried out to confirm that the system's features operated in accordance with the defined functional requirements and met user expectations (Pressman & Maxim, 2020). While unit testing focuses on individual components in isolation, functional testing evaluates the system as a whole by examining how different modules interact during realistic usage scenarios (Sommerville, 2016). During this phase, core system functionalities were tested, including user registration and authentication, profile management, workout logging, calorie burn calculation, daily caloric intake recommendations, and data visualization.

Each function was tested using both valid and invalid input values to ensure that the system responded correctly under normal and erroneous conditions, as recommended in functional testing standards (IEEE, 2018). For example, incorrect login credentials were intentionally submitted to verify that the system generated appropriate error messages and prevented unauthorized access. Similarly, incomplete or invalid workout entries were provided to confirm that input validation mechanisms effectively restricted incorrect data submissions. These tests ensured that the system enforced data accuracy and integrity during user interaction (Pressman & Maxim, 2020).

Functional testing also confirmed that data consistency was maintained across interconnected system components. Profile information provided during user registration was examined to

ensure it was correctly applied in calorie calculations and accurately reflected in progress summaries and dashboard analytics. This process validated proper data flow between system modules and eliminated discrepancies between user inputs and computed results (Sommerville, 2016). Overall, the results of functional testing demonstrated that the application satisfied its specified requirements and performed reliably from the user's perspective. This testing approach was essential in confirming that the system delivered its intended services and that the feedback and recommendations provided to users were accurate, consistent, and actionable (IEEE, 2018).

#### **4.4.3 Usability Testing**

Usability testing focused on evaluating how easily users could interact with the web application, CalorieAi, and successfully accomplish their intended tasks (ISO 9241-11, 2018). Since user adoption and continued usage are strongly influenced by system usability, this testing approach was considered essential to the overall success of the project (Nielsen, 2012). During usability testing, selected participants were asked to perform common tasks such as logging a workout, viewing calorie estimates, checking daily caloric requirements, and interpreting feedback provided by the system.

Observations were made regarding navigation flow, clarity of instructions, ease of data entry, and overall user satisfaction, which are key dimensions in usability evaluation (Rubin & Chisnell, 2008). Participants were also encouraged to verbalize their thoughts and challenges while interacting with the system, a technique commonly referred to as the think-aloud method (Nielsen, 2012). Structured questionnaires based on established usability criteria including effectiveness, efficiency, and user satisfaction were used to collect quantitative and qualitative feedback (ISO 9241-11, 2018).



The results highlighted areas where users experienced confusion or difficulty, such as unclear interface labels, unnecessary interaction steps, and insufficient feedback messages. These findings are consistent with prior research indicating that poor interface clarity can negatively affect user engagement and task completion (Rubin & Chisnell, 2008). Insights obtained from usability testing informed interface refinements and improvements to system clarity and responsiveness. This process ensured that the application was not only technically functional but also user-friendly and accessible to a broad range of users (Sommerville, 2016). By addressing usability concerns early, the system was refined to support a positive user experience, which is critical for sustained engagement and effectiveness in health-related applications (Nielsen, 2012).

#### **4.4.4 Acceptance Testing**

Acceptance testing was conducted to determine whether the developed system, CalorieAi, met the expectations of its intended users and satisfied the overall project objectives (Sommerville, 2016). This phase focused on validating the system from an end-user perspective rather than purely technical criteria, ensuring that the application delivered practical value in real usage contexts (Pressman & Maxim, 2020).

During acceptance testing, participants evaluated the system based on its perceived usefulness, accuracy, ease of use, and relevance to personal fitness goals. These evaluation criteria are widely recognized as key indicators of user acceptance and system success (Davis, 1989). Participants assessed whether the calorie estimates and daily caloric recommendations were understandable, realistic, and helpful in supporting informed decision-making related to health and fitness management (Westerterp, 2013).

Feedback was collected regarding overall user satisfaction and the perceived value of the system in supporting daily activity monitoring and calorie management. Acceptance testing

also examined whether the system aligned with the original project scope and functional requirements defined during the system design phase (Sommerville, 2016). Successful completion of this testing phase indicated that the application was suitable for deployment as a functional prototype and capable of supporting real-world use scenarios (Pressman & Maxim, 2020). The outcomes of acceptance testing demonstrated that the system achieved its intended purpose and was ready for academic evaluation and potential future enhancement.

#### **4.4.5 Selected Testing Approach**

While multiple testing approaches were applied in this study, functional testing and usability testing were selected as the primary testing strategies for CalorieAi. This decision was informed by the project's emphasis on delivering accurate health-related computations and ensuring a positive user experience, which are widely recognized as critical success factors in health and fitness applications (Pressman & Maxim, 2020; ISO 9241-11, 2018). Functional testing ensured that the system's core features operated correctly and consistently in accordance with specified requirements, while usability testing confirmed that users could interact with the system effectively without technical difficulty (Sommerville, 2016).

The integration of these testing approaches enabled a balanced evaluation of both system performance and user satisfaction, which is recommended for interactive web-based systems (Nielsen, 2012). The testing and evaluation of the AI-powered web application revealed that the system effectively met its intended objectives and functional requirements. Results from unit and functional testing confirmed that the core system components including user authentication, workout logging, energy expenditure calculation, and daily caloric needs estimation operated accurately and consistently (Pressman & Maxim, 2020). The implementation of standardized metabolic formulas and rule-based AI logic produced reliable

outputs that aligned with expected theoretical values, indicating a high level of computational correctness (Mifflin et al., 1990; Ainsworth et al., 2011).

Findings from usability testing showed that users were able to navigate the system with minimal difficulty and complete key tasks, such as logging workouts and viewing caloric recommendations, efficiently. Participants reported that the interface was intuitive, the feedback provided was clear, and the visual presentation of results enhanced their understanding of personal fitness data, which aligns with established usability principles for health information systems (ISO 9241-11, 2018; Nielsen, 2012). Observational results further indicated increased user confidence in interacting with the system as familiarity improved, suggesting effective learnability.

Acceptance testing demonstrated that users found the system relevant, useful, and appropriate for supporting fitness and weight management goals, consistent with established user acceptance models (Davis, 1989). Most participants expressed satisfaction with the accuracy of calorie estimations and the practical guidance provided on calorie deficit, maintenance, and surplus. Overall, the findings indicate that the developed system is functional, user-friendly, and suitable for deployment as a reliable prototype, with strong potential for further enhancement and real-world application (Sommerville, 2016).

#### **4.5 Implementation of the Current System**

System implementation refers to the process of deploying a developed system like CalorieAi into an operational environment where it can be accessed and used by intended users (Laudon & Laudon, 2020). The choice of implementation strategy is critical, as it determines how smoothly users transition from existing methods to the new system, and how risks such as system failure, data loss, or user resistance are managed (Pressman & Maxim, 2020).

In the context of this study, which involves an AI-powered web application for monitoring energy expenditure and daily caloric needs, implementation strategies were carefully considered to ensure minimal disruption, accurate system evaluation, and user acceptance (Sommerville, 2016). Several implementation strategies are commonly applied in information systems development, including parallel implementation, pilot implementation, direct implementation, and phased implementation (Stair & Reynolds, 2018). Each strategy offers distinct advantages and limitations depending on system complexity, size of the user base, and organizational context (Laudon & Laudon, 2020). These strategies are discussed below in relation to the proposed system.

#### **4.5.1 Parallel implementation**

Parallel implementation involves running the newly developed system, CalorieAi, alongside the existing system for a specified period (Stair & Reynolds, 2018). During this phase, both systems operate simultaneously, allowing users to compare outputs and verify the accuracy of the new system before fully transitioning to it (Laudon & Laudon, 2020). This strategy is widely regarded as one of the safest implementation approaches, as it minimizes risk by ensuring that users can revert to the old system if issues arise (Pressman & Maxim, 2020).

In the context of the AI-powered web application, parallel implementation would involve users continuing to rely on traditional calorie estimation methods or existing fitness applications while also using the developed system to log workouts and calculate caloric needs. The outputs from both methods can then be compared to assess accuracy, consistency, and usability (Sommerville, 2016). The major advantage of parallel implementation is risk reduction. Since the existing system remains operational, users are not entirely dependent on the new system during the initial deployment phase (Laudon & Laudon, 2020). However, this approach may require additional effort from users, as they must interact with two systems simultaneously

(Stair & Reynolds, 2018). Despite this limitation, parallel implementation is valuable for validating system reliability, especially for health-related applications where accuracy is critical (Pressman & Maxim, 2020).

#### **4.5.2 Pilot implementation**

Pilot implementation involves deploying the system to a small, selected group of users before full-scale rollout (Stair & Reynolds, 2018). This approach allows developers to observe system performance, identify operational issues, and gather feedback from real users in a controlled environment (Laudon & Laudon, 2020). Pilot implementation is particularly useful for evaluating usability, user acceptance, and overall system effectiveness under realistic usage conditions (Sommerville, 2016).

For this project, pilot implementation was considered an appropriate strategy due to the academic and prototype nature of the system. A selected group of University of Professional Studies, Accra (UPSA) students who regularly engage in physical activity would use the system over a defined period. Their feedback on system functionality, interface design, and calorie estimation accuracy would be collected and analyzed to guide further system refinement (Pressman & Maxim, 2020).

The primary advantage of pilot implementation is its ability to uncover issues that may not be evident during laboratory-based testing phases (Sommerville, 2016). It also promotes user involvement and acceptance by allowing early adopters to contribute directly to system improvement (Laudon & Laudon, 2020). However, pilot implementation does not fully represent the challenges associated with large-scale deployment, such as high concurrent usage and infrastructure stress (Stair & Reynolds, 2018). Nonetheless, it remains a highly effective

strategy for validating prototype systems and ensuring readiness for wider use, particularly in academic and health-related applications (Pressman & Maxim, 2020).

#### **4.5.3 Direct implementation**

Direct implementation, also referred to as the “big bang” approach, involves completely replacing an existing system with a new system at a specific point in time, after which users immediately transition to the new solution without fallback options (Stair & Reynolds, 2018). This strategy is often adopted when the existing system is outdated, ineffective, or no longer sustainable for continued operation (Laudon & Laudon, 2020).

Applying direct implementation to the AI-powered web application, CalorieAi, would require users to discontinue traditional calorie tracking methods or existing fitness applications and rely solely on the developed system for monitoring energy expenditure and daily caloric needs. While this approach enables immediate adoption and eliminates the cost and complexity associated with operating multiple systems concurrently, it carries significant implementation risk (Pressman & Maxim, 2020).

The major disadvantage of direct implementation is the absence of a safety net. If system errors occur or users experience usability challenges, there is no alternative system available for continuity of operations (Sommerville, 2016). This risk is particularly critical for health-related applications, where inaccuracies or system failures may negatively influence user decisions and trust (Laudon & Laudon, 2020). Given the prototype nature of this project and the importance of ensuring accuracy, reliability, and user confidence, direct implementation is considered less suitable at the current stage of development. However, it may become appropriate in future iterations after extensive testing, validation, and user acceptance have been achieved (Pressman & Maxim, 2020).

#### **4.5.4 Phased Implementation**

Phased implementation involves introducing a new system gradually, either by deploying system functionalities in stages or by rolling out the system to different user groups over a period of time (Laudon & Laudon, 2020). This approach reduces implementation risk by allowing users and developers to concentrate on specific system components before full deployment, thereby minimizing the impact of potential errors or failures (Stair & Reynolds, 2018).

For the proposed AI-powered web application, phased implementation could involve initially deploying core functionalities such as user registration, authentication, and workout logging, followed by the integration of calorie estimation, daily caloric requirement computation, AI-driven recommendations, and data visualization modules. This staged deployment enables each system component to be tested, evaluated, and refined independently before subsequent features are introduced (Sommerville, 2016).

A key advantage of phased implementation is its flexibility and reduced risk of system failure. Users are able to adapt progressively to new functionalities, while developers can identify and resolve issues incrementally without disrupting the entire system (Pressman & Maxim, 2020). Additionally, this approach supports continuous feedback and improvement, which is particularly valuable for user-centered and data-driven applications. However, phased implementation may extend the overall deployment timeline due to the sequential release of features. Despite this limitation, it is widely regarded as a suitable strategy for complex systems and aligns closely with agile development principles, which emphasize incremental delivery, continuous testing, and iterative improvement (Beck et al., 2001).

## 4.6 System Documentation

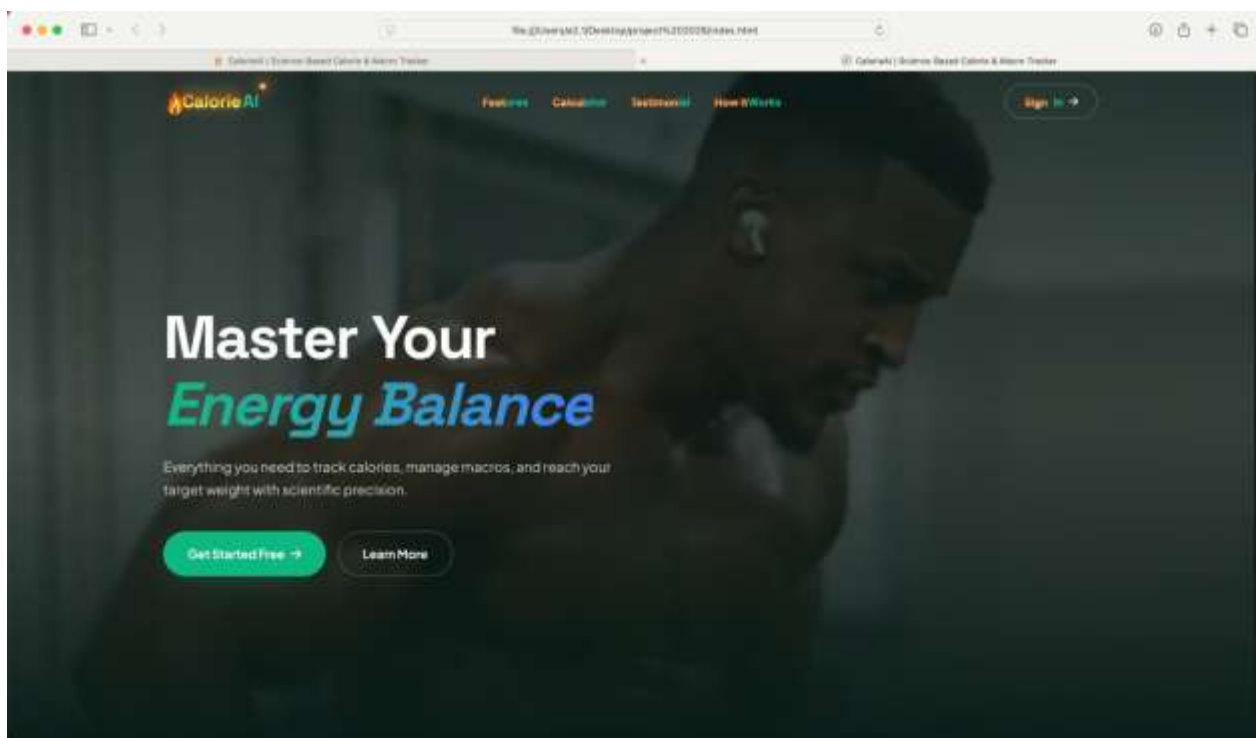
### 1. Public Landing & Marketing Interface

The landing page serves as the public facade of CalorieAI. It utilizes a dynamic, responsive layout to introduce the application's scientific foundation. The interface includes:

**Hero Section:** A visually striking introduction that immediately sets a professional, fitness-oriented tone.

**Scientific Pillars:** A dedicated section explaining the clinical approach behind the app (Mifflin-St Jeor Equation, MET tracking).

**Interactive BMR Calculator:** A live, Javascript-powered utility allowing unregistered users to input their height, weight, and age to get an immediate preview of their Basal Metabolic Rate (BMR) before signing up. This drives user engagement immediately.





## — OUR SCIENTIFIC FOUNDATION

# Built on Clinical Data, Not Guesswork.

03  
ACTIVE PROTOCOLS



### MIFFLIN-ST JEOR EQUATION

The most used equation for calculating basal metabolic rate (BMR) was used to determine your body's most energy needs at rest with 80% accuracy.

— APPROXIMATELY 80% ACCURATE



### MET-BASED TRACKING

Metabolic equivalent of task (MET) allows us to quantify the intensity of your workouts, helping us estimate accurately from rest to working heart rate.

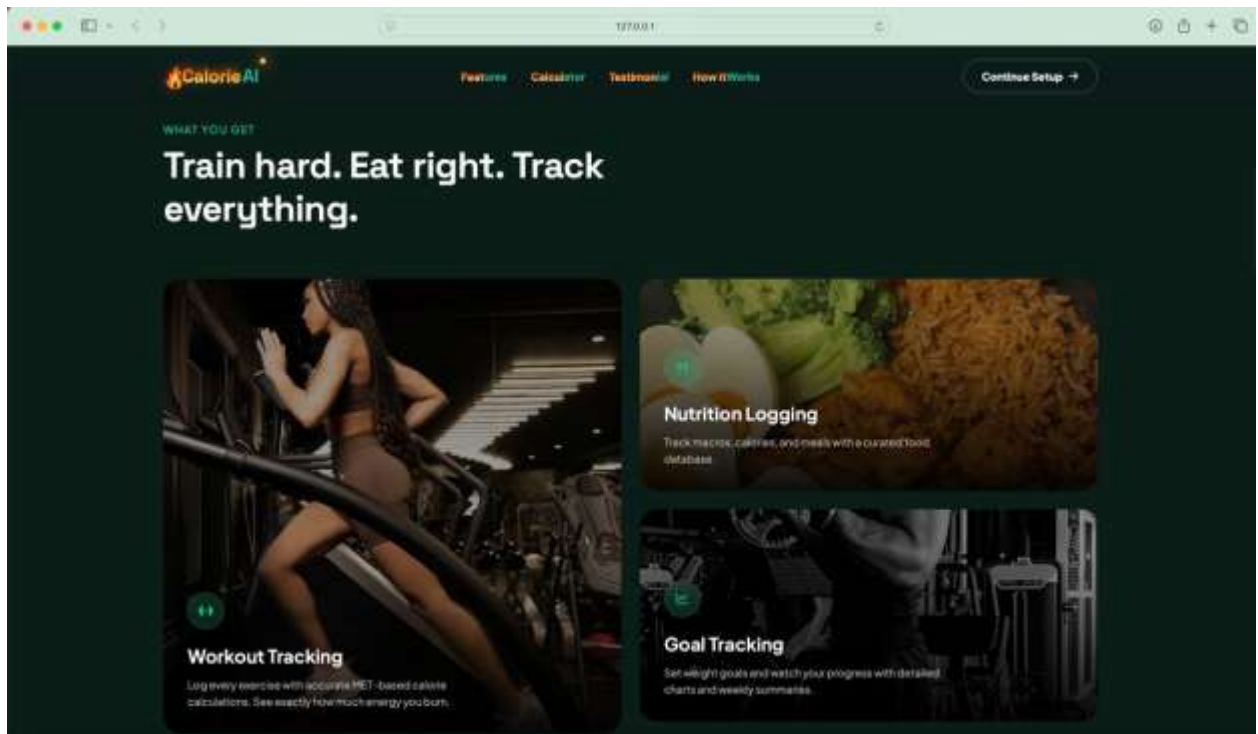
— APPROXIMATELY 80% ACCURATE



### ADAPTIVE AI HEURISTICS

Our app engine analyzes your energy output in real time, it suggests the deficit for the next 24 hrs. Adjusted for your muscle gain based on your trends.

— APPROXIMATELY 80% ACCURATE



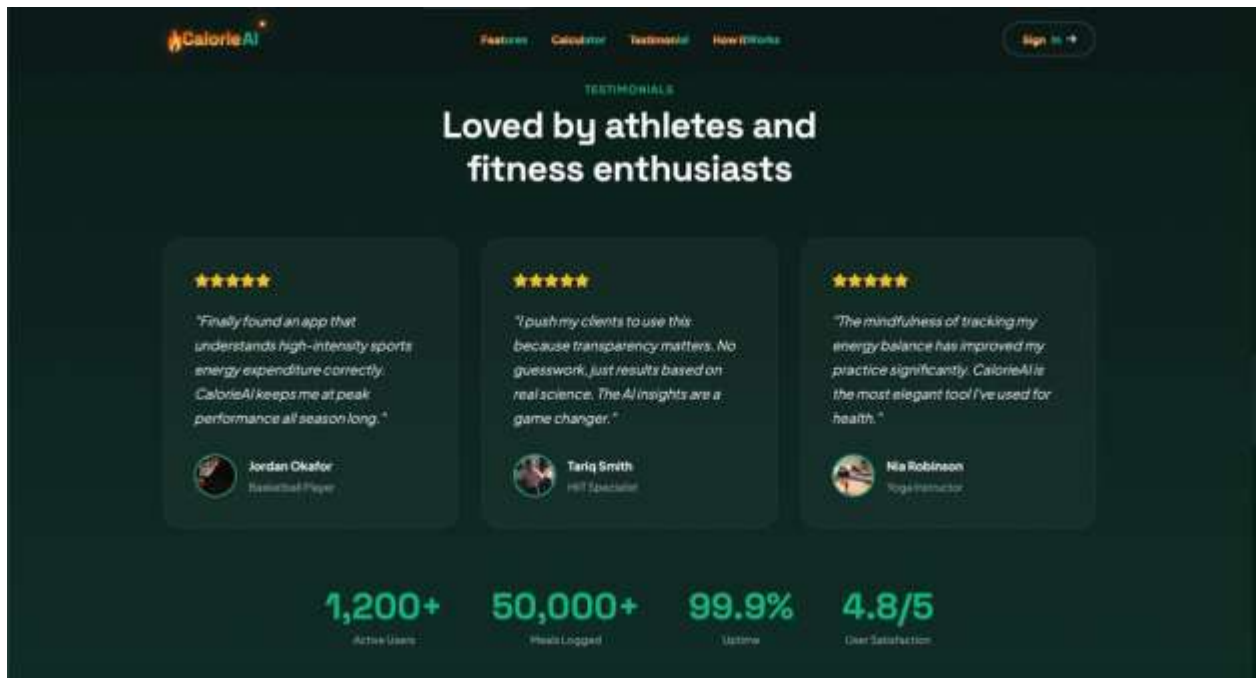
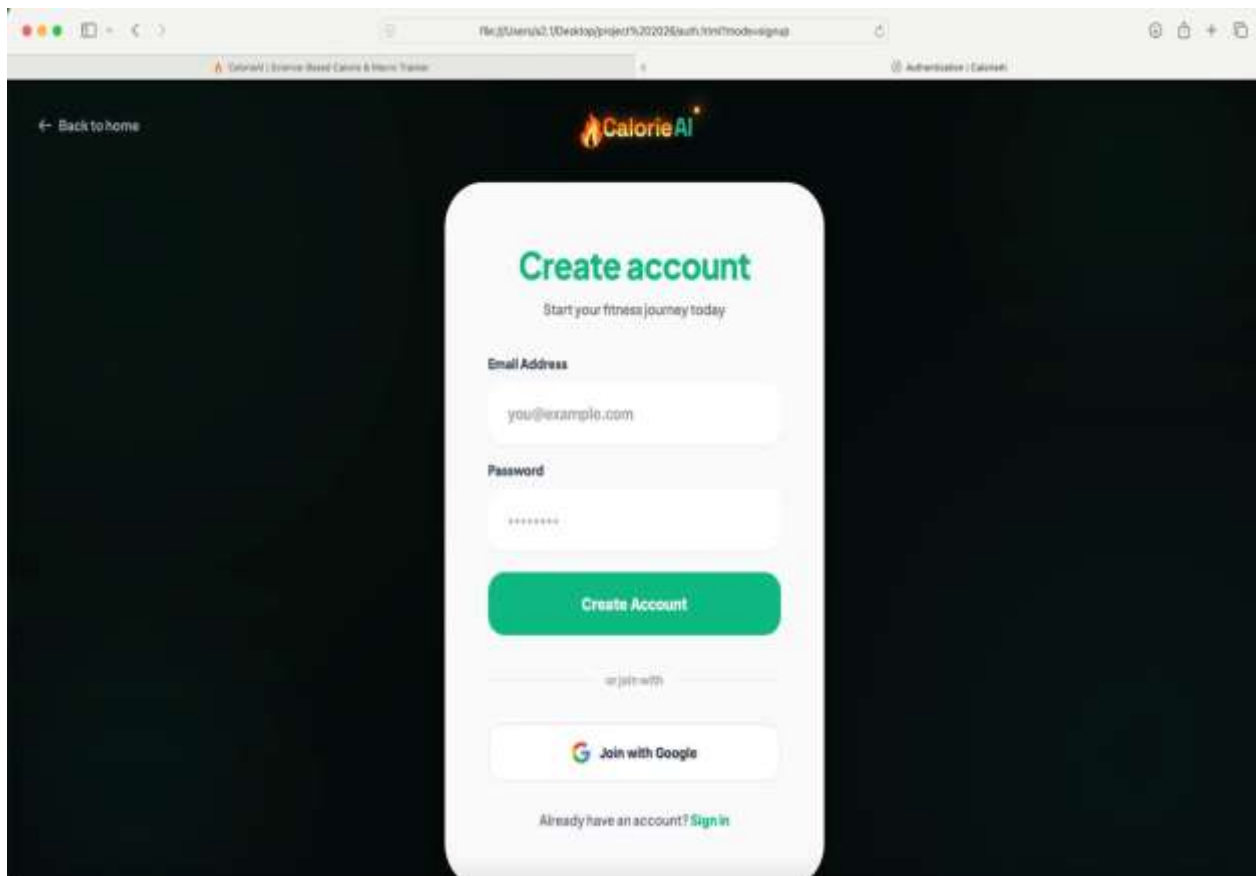
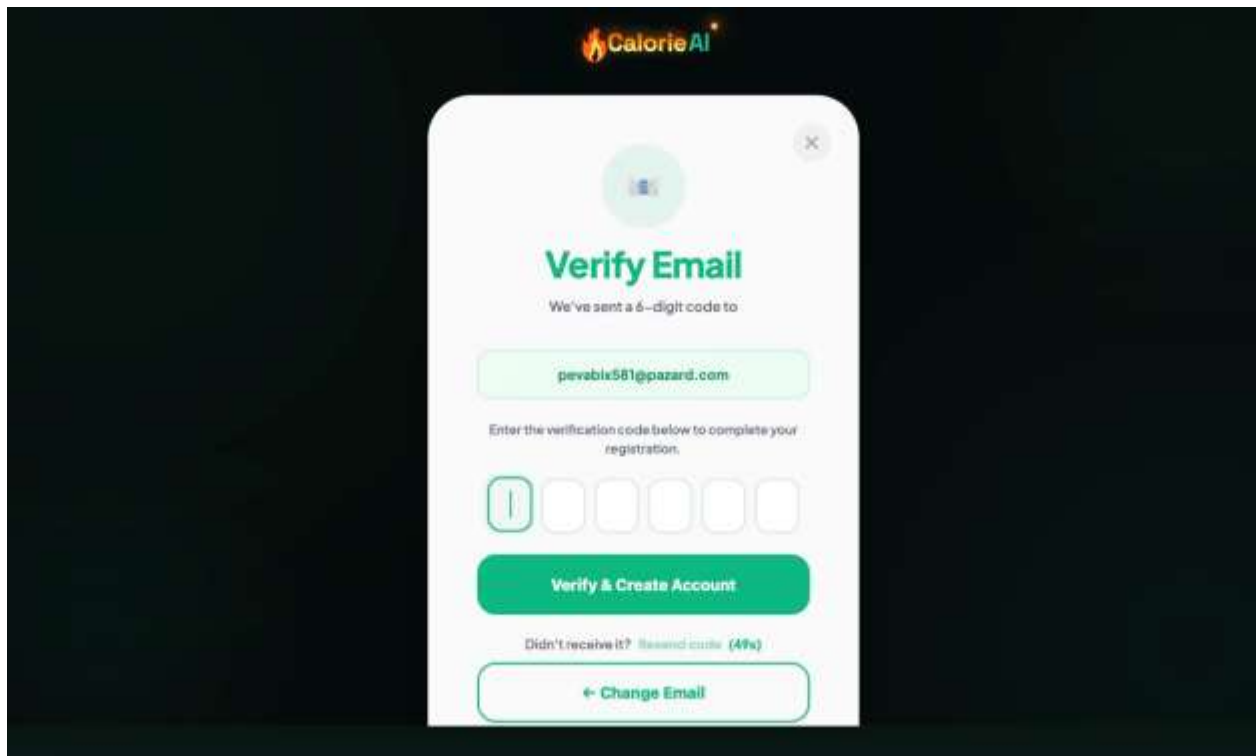


Figure 6: Public Landing & Marketing Interface

## System Authentication page

To protect user health data natively, the system utilizes Google's Firebase Authentication. This interface allows users to create an account or sign in securely. The form handles state management and error validation on the client side, ensuring no unauthorized users gain access to the data dashboard





**Figure 7: System Auth/Create Account Page**

### **System Onboarding Page:**

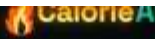
Once an account is created, the system triggers a mandatory, linear onboarding flow designed to calculate the precise physiological blueprint of the user.

Step 1: Basic Info : Captures essential demographic data.

Step 2: Measurements : Captures current weight and height metrics.

Step 3: Goal Definition : Users interact with a dynamic selector to choose between "Lose", "Maintain", or "Gain". The system executes the Mifflin-St Jeor equation in real-time to display the exact caloric deficit or surplus required. For weight loss/gain, a sub-menu expands allowing the user to select the "pace" (Slow, Moderate, Fast).

Step 4: Activity Rate : Factors in the user's daily lifestyle multiplier to finalize their personalized Daily Energy Expenditure. All of this state is then securely pushed to the cloud database (Firestore).





### Let's get to know you

This helps us calculate your metabolic rate accurately

Full Name

Date of Birth

Gender

☒ Male ☐ Female

Next →





### Your measurements

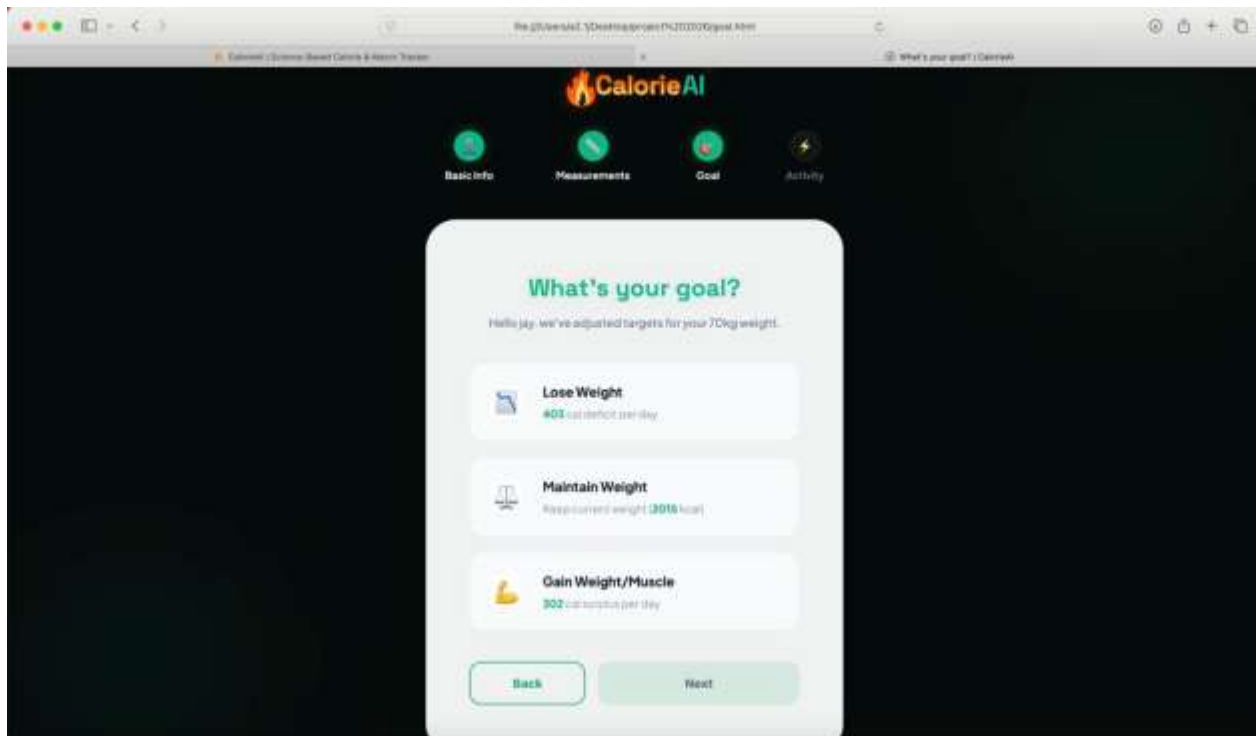
We use this to calculate your calories needed

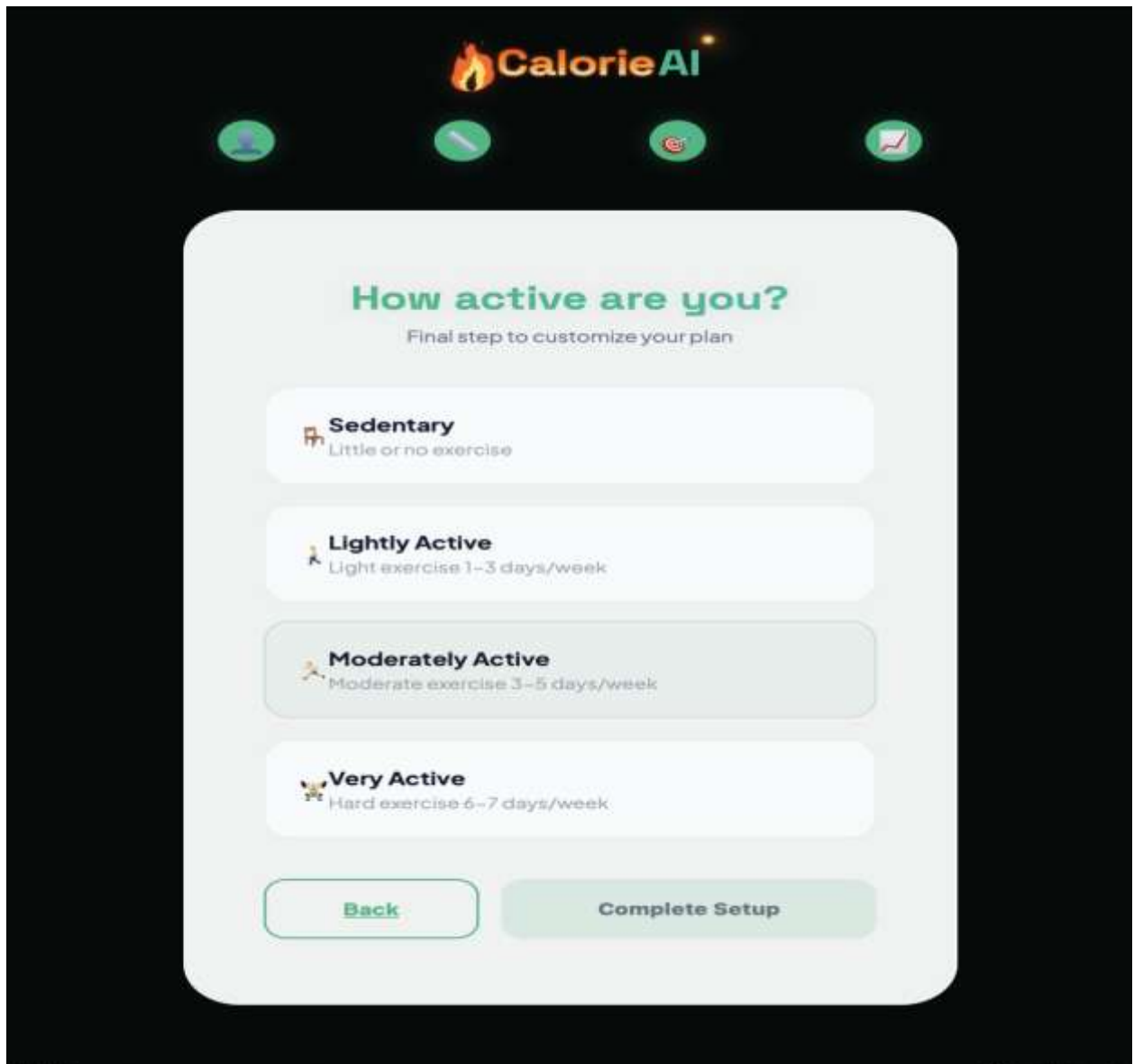
Height (cm)

Weight (kg)

← Back

Next →





**Figure 8: System Onboarding Page**

### **System Dashboard page.**

The CalorieAi Dashboard serves as the user's main control center, combining all tracking, logging, and analytics features into a single, easy-to-navigate interface used daily after login. At the very top, a personalized welcome header greets the user by name and displays the current date, immediately orienting them to their daily session. Directly below, the Log Food and Log Workout action buttons are placed prominently as the first interactive elements on the page.



Log Food opens a modal where users can record a meal by searching a built-in food database (which includes local Ghanaian dishes), entering a custom meal manually, or using the AI-powered food scanner which allows the user to upload or take a photo of their plate and have the Google Gemini Vision API automatically identify the food and estimate its full nutritional breakdown including calories, protein, carbs, and fat. Log Workout opens a detailed logging panel where users can select an activity type, set the duration, choose an intensity level, and have the system calculate and record their active calorie burn in real time.

Following the action buttons, the Stats Grid displays four key daily metric cards. The Daily Target card shows the user's personalized recommended calorie intake calculated from their profile goals. The BMR card displays their Basal Metabolic Rate—the number of calories their body burns at complete rest—serving as a reference point for their energy needs. The Today Burned card tracks the total active calories the user has burned from all workouts logged that day. The Weekly Workouts card counts the total number of exercise sessions the user has completed during the current week, encouraging consistency. Below the stats, the Today's Macros Bar provides a live snapshot of the user's macronutrient consumption for the day, showing the total grams of protein, carbohydrates, and fat logged so far, allowing users to immediately assess the nutritional quality of their intake, not just the calorie count.

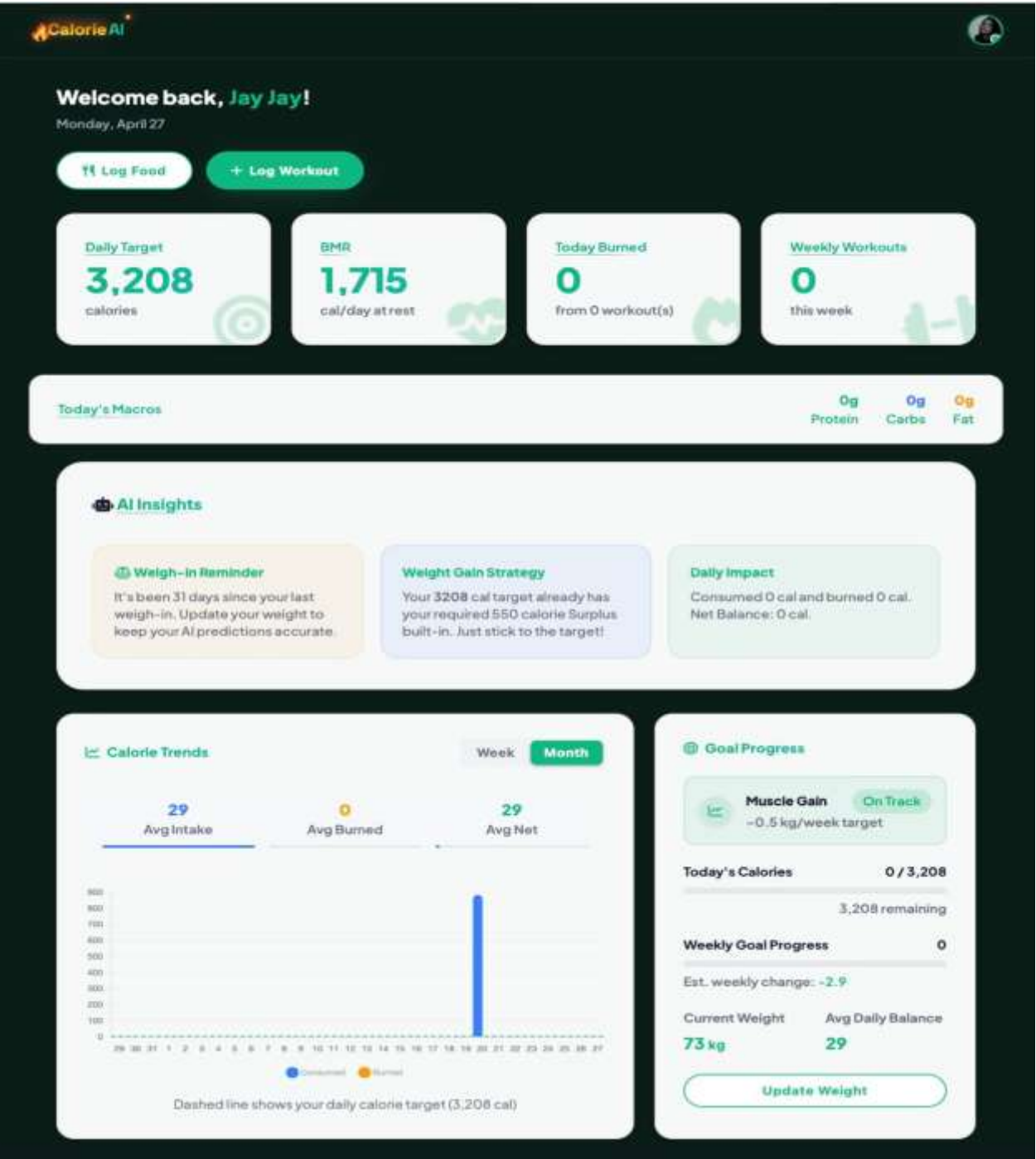
The AI Insights section then delivers personalized, data-driven coaching feedback based on the user's unique patterns. It analyzes the user's logged meals, workout activity, and progress data to surface actionable alerts—for example, flagging consistently low protein intake, recognizing streaks of hitting calorie targets, or advising on balancing calorie deficit and activity levels. This section acts as an intelligent virtual coach, ensuring users receive meaningful guidance while they are still actively engaged with their dashboard.

The main Analytics Grid presents two panels side by side. On the left, the Calorie Trends chart plots the user's daily caloric intake and calories burned across the selected time period (past week or past month), with their daily calorie target shown as a reference line—making it easy to spot patterns, identify problem days, and understand long-term dietary behavior. On the right, the Goal Progress panel tracks real-time adherence to the user's fitness goal. It displays a progress bar for today's calorie consumption versus their daily target, a weekly goal progress bar showing estimated adherence over the full week, and a projected weekly weight change based on the current calorie balance giving the user a clear picture of whether they are on track to meet their goal.

Further down, the Weight Progress section allows users to log their body weight and view an interactive Chart.js graph plotting their weight trend over a chosen period of one week, one month, or three months. Summary statistics show the user's current weight, the change over the selected period, and the total change since they began tracking, making it easy to correlate dietary habits and workout consistency with real physical results over time.

Finally, the Weekly Summary sits at the bottom of the dashboard as a comprehensive retrospective report. It aggregates the week's total calories consumed and burned compared to the previous week, the total number of workouts completed, and the net weight change for the week. It also tracks how many days the user hit their daily calorie target, breaks down average daily macro intake for protein, carbs, and fat, displays total active workout minutes logged, and awards achievements earned during the week such as Workout Warrior or Protein Power—providing a full accountability review without interrupting the daily workflow. Overall, the dashboard is structured to prioritize action first, immediate context second,

intelligent feedback third, detailed analytics fourth, and retrospective review last—enabling users to efficiently manage their nutrition, fitness, and long-term progress all in one place.





## Log Workout

Push your limits and track your progress

### CHOOSE ACTIVITY

|                                                                                                 |                                                                                             |                                                                                             |                                                                                                |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <br>Gym        | <br>Yoga   | <br>HIIT   | <br>Dancing |
| <br>Basketball | <br>Soccer | <br>Tennis | <br>Boxing  |

### DURATION MODE

|                                                                                            |                                                                                               |                                                                                             |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|  Manual |  Stopwatch |  Timer |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

### HOURS

### MINUTES

### INTENSITY LEVEL

|                                                                                           |                                                                                              |                                                                                               |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
|  Light |  Moderate |  Vigorous |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|



## Log Food

Track your nutrition and stay on target

### SELECT MEAL TYPE



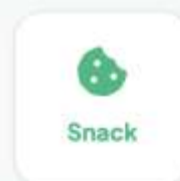
Breakfast



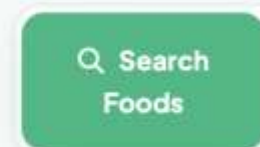
Lunch



Dinner



Snack



 Custom  
Entry

 Scan Plate  
**AI**

Search for foods...

**Waakye (with Fish, Egg & Wele)**

1 bowl

 40g  95g  28g

**850**

CAL

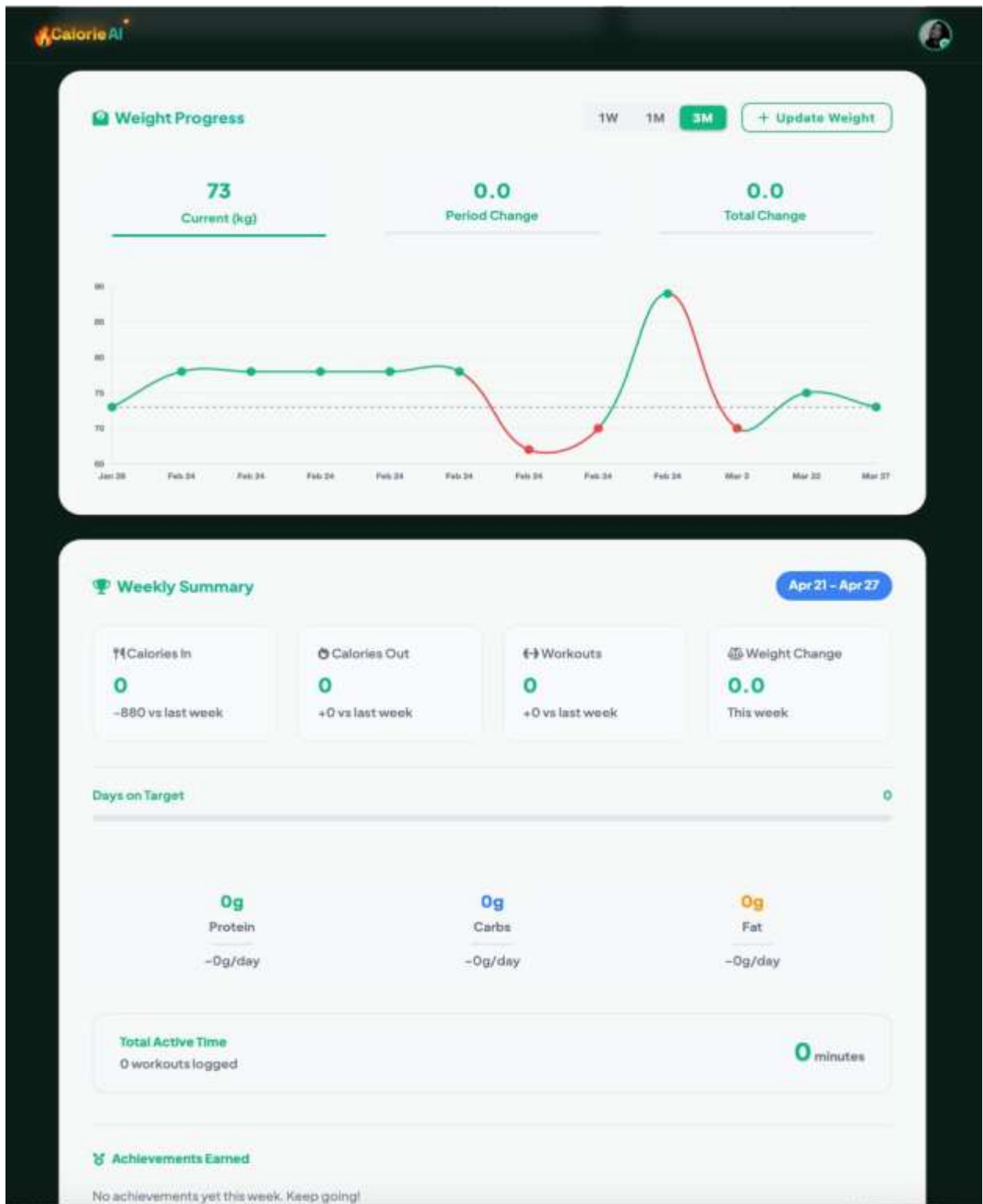


Figure 9: System Dashboard Page

## System Update Profile Page

This feature serves as the centralized hub for managing personal user data and account settings.

In a fitness application, physiological metrics are the foundation of accurate tracking.

The Update Profile section allows users to:

Modify core personal attributes such as their current age, height, weight, and baseline activity level.

Adjust their overarching fitness goals (e.g., shifting from a "weight loss" phase to a "muscle building" phase).

When a user updates these details, the application instantly recalculates their daily target caloric intake and macronutrient splits, ensuring that their dashboard goals always remain perfectly aligned with their current physical state and objectives.

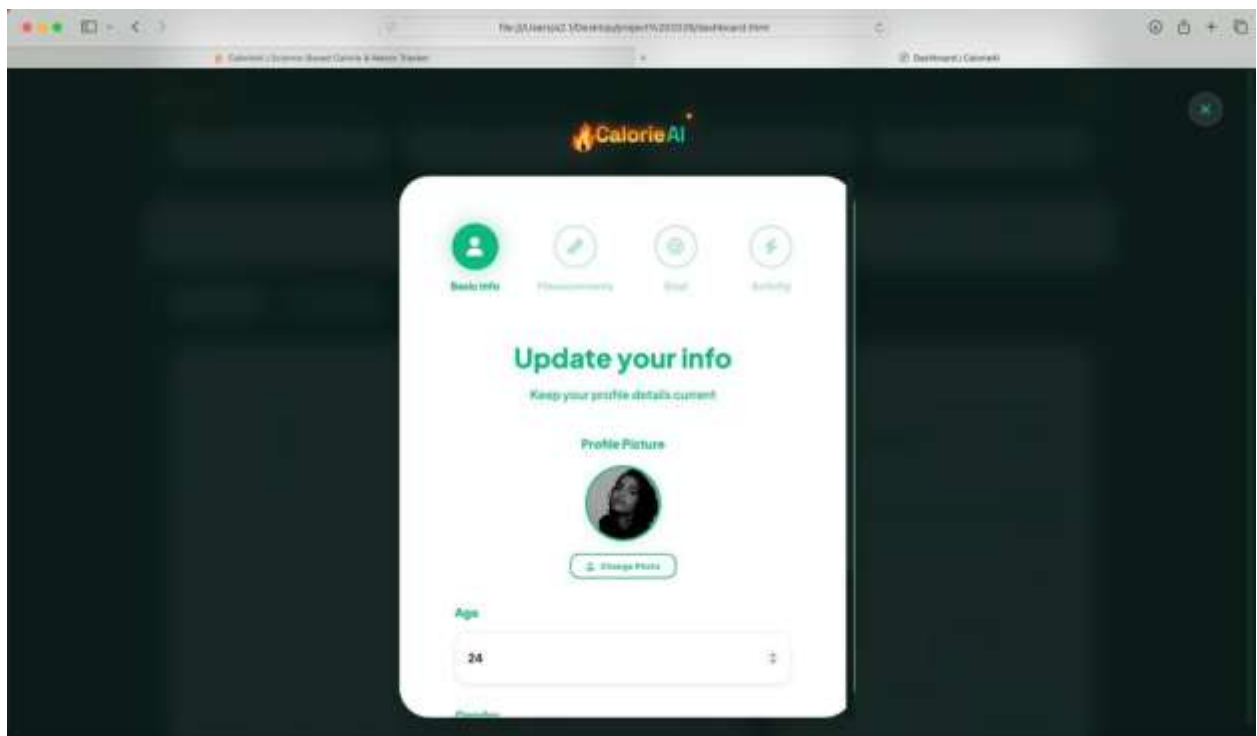


Figure 10: System User Update Info

## Workout History

The Workout History provides a comprehensive, chronological archive of all previously recorded physical activities. Instead of just seeing today's active calorie burn, this feature allows users to:

Scroll back through past days, weeks, or months to review their historical exercise routines.

Track their physical consistency and pinpoint periods of high or low activity.

Review specific workout details (like duration and estimated calories burned) to analyze performance trends over time, helping users stay accountable to their long-term fitness regimens.

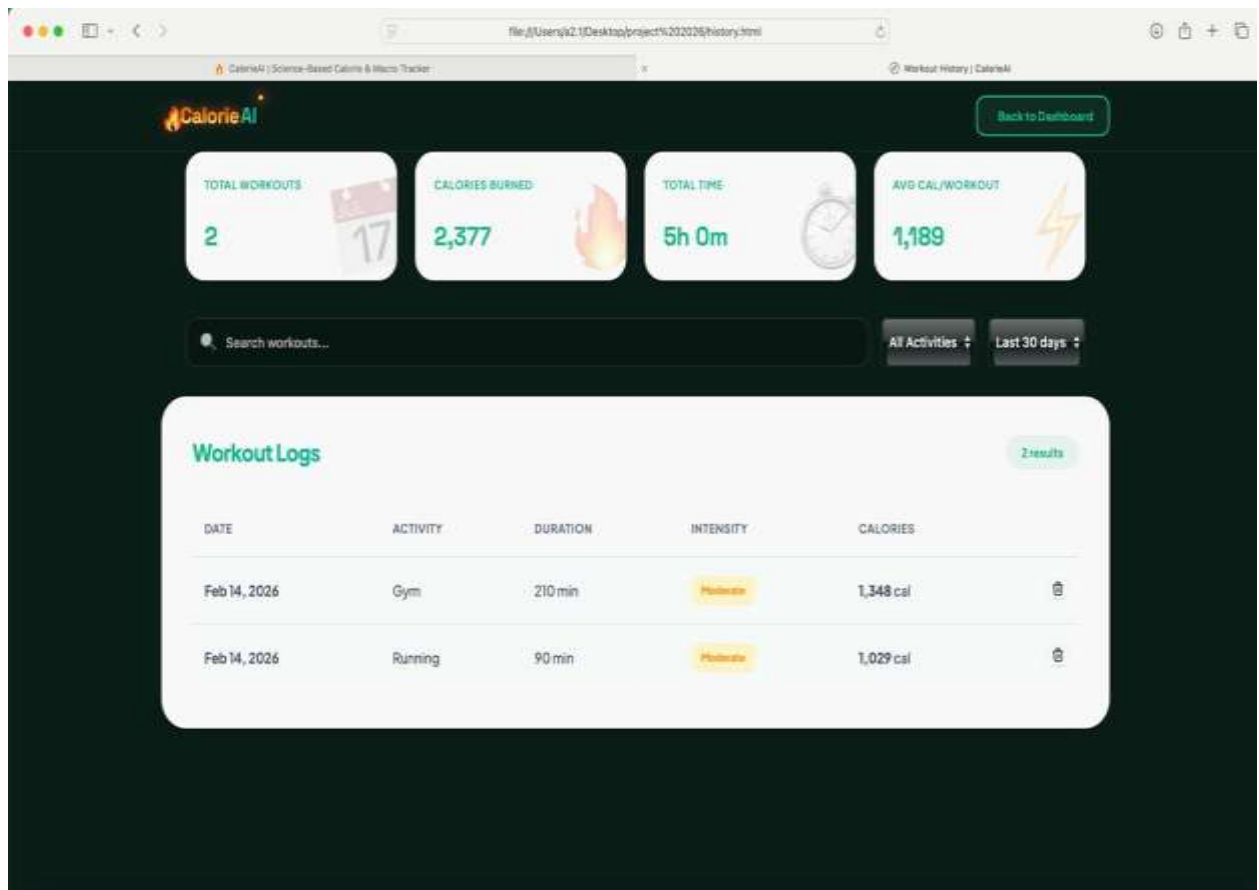


Figure 11: System Workout History Page



## Meal History page

Serving as a detailed digital food diary, the Meal History archives every single meal and snack the user has logged, including those processed by the AI food scanner. This feature is crucial for deep nutritional reflection because it enables users to:

Look back at specific dates to audit their exact dietary choices and review the complete breakdown of Calories, Protein, Carbs, and Fats for past meals.

Identify recurring eating patterns or triggers such as noticing which specific meals consistently push them over their daily calorie limit, or which breakfasts yield the best energy levels for their workouts.

Retrieve and reuse past AI-scanned data, making it easier to reflect on what worked during successful weeks on their health journey

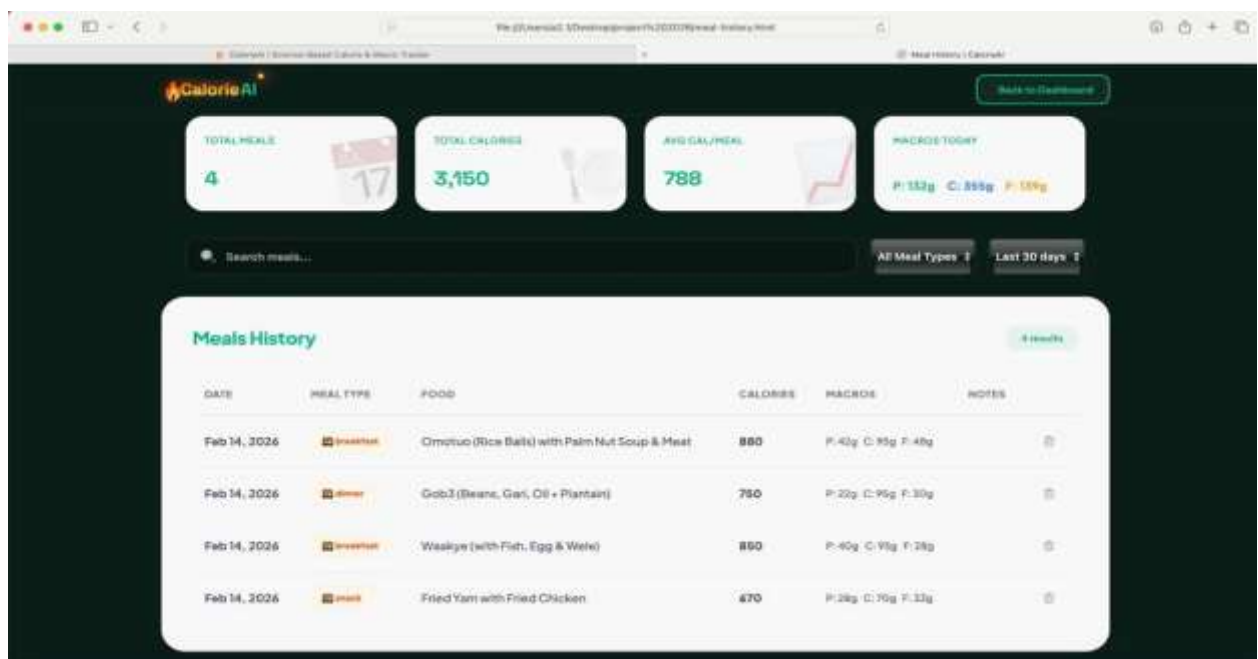


Figure 12: System Meal History Page

## 4.7 Implementation Challenges

The development and implementation of the AI-powered web application, CalorieAi, presented several challenges that required practical solutions. One key challenge was ensuring accurate calorie computation, as incomplete or inconsistent user inputs affected calculation

results. This issue was addressed by implementing a robust client-side input validation mechanism. Before processing data, the system ensures fields are not empty, applies numeric validation to prevent "NaN" errors for biometrics, and enforces range validation. Once validated, JavaScript executes the calorie calculation algorithm locally and dynamically renders results without requiring a page reload (Pressman & Maxim, 2020).

Additional technical challenges emerged during the integration of the Generative AI food scanner. The external API occasionally returned Markdown-formatted responses instead of pure JSON, causing application crashes. This was resolved by implementing robust string parsing and regular expression (RegEx) cleanup scripts (Chen & Zhang, 2020). Furthermore, maintaining accurate active workout timers across accidental page reloads was mitigated by implementing a state-persistence mechanism utilizing the browser's localStorage API (Sommerville, 2016).

Finally, managing user profile image uploads without relying on external cloud storage buckets was addressed by converting images to Base64 strings using the FileReader API and storing them directly within the Firestore user document (Fielding & Taylor, 2017). To ensure documents did not exceed Firestore's 1MB limit, client-side image compression was considered to manage the Base64 string length effectively. Variations in user understanding of workout intensity and caloric feedback also influenced system interaction. To address this, explanatory labels and tooltips were incorporated to improve clarity and reduce user confusion, consistent with established usability and human-computer interaction principles (Nielsen, 2012; ISO 9241-11, 2018).

Early usability testing further revealed navigation challenges, which were resolved through iterative interface refinements, simplified menu structures, and improved visual cues, aligning with user-centered design practices (Sharp, Rogers, & Preece, 2019). From an implementation perspective, integration challenges between frontend components and Firebase backend

services, including data synchronization delays, were resolved by optimizing database queries and implementing real-time listeners, as recommended for cloud-based web systems (Buyya et al., 2019). Testing constraints related to time and sample size were addressed through focused pilot testing with representative users, an approach commonly adopted in academic software development projects (Sommerville, 2016). Overall, addressing these challenges contributed to improved system stability, usability, and reliability for CalorieAi.

#### **4.8 Chapter Summary**

This chapter focused on the implementation, testing, and evaluation of the AI-powered web application, CalorieAi, developed to monitor energy expenditure and daily caloric needs. It provided an overview of the testing approaches employed, including unit testing, functional testing, usability testing, and acceptance testing, highlighting how each contributed to ensuring system reliability and user readiness (Pressman & Maxim, 2020). The chapter also discussed the implementation strategies considered for deploying the system, emphasizing pilot and phased implementation as the most suitable approaches for a prototype health-related application (Laudon & Laudon, 2020).

These strategies ensured reduced risk, user involvement, and gradual system refinement (Sommerville, 2016). Furthermore, the chapter examined challenges encountered during testing and implementation, along with the solutions adopted to overcome them. Addressing these challenges improved system accuracy, usability, integration efficiency, and overall performance. In conclusion, the activities described in this chapter demonstrate that the developed system, CalorieAi, was implemented systematically, tested rigorously, and refined effectively. The outcomes confirm that the application meets its intended objectives and is suitable for deployment as a functional prototype. The next chapter will present the overall summary of the study, conclusions drawn, and recommendations for future enhancement.

## **CHAPTER FIVE**

### **SUMMARY, CONCLUSIONS AND RECOMMENDATIONS**

#### **5.1 Introduction**

This chapter closes the study by summarizing the design and development of an AI-based web application, CalorieAi, that tracks energy expenditure and estimates daily calorie needs. This section restates the study aims and the research problem, defines the scope and limits of the work, suggests directions for future studies and system changes, and closes the project. This chapter presents the study results and discusses how they contribute to research on information technology and digital health solutions (Chen & Zhang, 2020).

#### **5.2 Summary**

This study was conducted to address ongoing challenges in accurately estimating energy expenditure and determining appropriate daily calorie intake for weight management. Many existing fitness applications rely on generalized formulas and limited personalization, which may reduce accuracy and user trust (Krebs & Duncan, 2015; Westerterp, 2013). To address this gap, an AI-based web application, CalorieAi, was developed to track physical activity and estimate daily caloric needs using validated metabolic models.

Basal Metabolic Rate (BMR) was calculated using the Mifflin–St Jeor equation, recognized for its predictive accuracy in adults (Mifflin et al., 1990; Frankenfield et al., 2005), while exercise energy expenditure was estimated using standardized MET values (Ainsworth et al., 2011). A rule-based AI approach was applied to ensure transparency and consistency in calorie recommendations (Russell & Norvig, 2021). The system was developed using modern web technologies and evaluated through structured software testing methods to confirm functionality, usability, and reliability (Sommerville, 2016). Results indicated that CalorieAi

produced consistent outputs, was easy to use, and effectively supported informed decision-making regarding calorie deficit, maintenance, or surplus.

### **5.3 Practical Implications**

The findings of this study carry significant implications for individuals, healthcare practitioners, institutions, and the digital health ecosystem. At the individual level, CalorieAi demonstrates that an AI-powered web application can provide accurate, actionable energy expenditure data without requiring expensive wearable hardware (Fielding & Taylor, 2017). By utilizing validated models like the Mifflin–St Jeor equation and standardized MET values, the system empowers users to make scientifically grounded decisions regarding caloric deficits, surpluses, or maintenance for effective weight management (Mifflin et al., 1990; Ainsworth et al., 2011; Hall et al., 2012). Furthermore, the AI-powered plate scanning feature addresses a primary barrier to engagement by automating the food logging process, significantly reducing the manual effort typically required for dietary tracking (Krebs & Duncan, 2015; Esteva et al., 2019).

For healthcare practitioners, CalorieAi serves as a reliable supplementary resource for preventive counseling. By providing a platform that generates transparent, interpretable data, practitioners can move beyond potentially biased patient self-reports toward a data-informed approach to nutritional planning (Prince et al., 2008; Westerterp, 2013). The system's web-based accessibility is particularly relevant for resource-constrained settings, allowing professionals to recommend a high-quality monitoring tool without imposing hardware costs on patients (World Health Organization, 2022).

At the institutional level, specifically within the University of Professional Studies, Accra (UPSA), this study highlights the value of integrating student-led digital health innovation into campus wellness programs. Given the rise of diet-related non-communicable diseases in urban

Ghana, CalorieAi provides a scalable, low-cost model for promoting nutritional awareness among students and staff (Bosu, 2015; World Health Organization, 2022). This project also underscores the success of interdisciplinary collaboration between IT and health science departments in addressing community-specific health challenges (Pressman & Maxim, 2020).

For the broader digital health industry, the study demonstrates how hybrid AI techniques combining rule-based reasoning with Generative AI can improve both computational accuracy and user engagement (Chen & Zhang, 2020; Russell & Norvig, 2021). The serverless architecture using Firebase and Google Gemini offers a modern, scalable development template that ensures real-time responsiveness and data security (Google Cloud, 2023; Bass, Clements, & Kazman, 2012). Ultimately, from a societal perspective, CalorieAi aligns with World Health Organization priorities by democratizing health information and empowering individuals to take ownership of their wellness outcomes through accessible technology (World Health Organization, 2020).

#### **5.4 Recommendations for Future Research**

Future research can expand the capabilities of CalorieAi and evaluate it across broader contexts. Integrating wearable sensors such as heart rate monitors and fitness trackers could enhance the accuracy of energy expenditure estimation, as physiological measures improve activity-based calorie assessment (Ainsworth et al., 2011; Fuller et al., 2020). Further studies may also explore adaptive machine learning techniques that learn from user behavior and historical data to provide increasingly personalized and predictive recommendations (Goodfellow, Bengio, & Courville, 2016).

Large-scale deployment involving diverse participant groups is recommended to assess generalizability and real-world performance (Creswell & Creswell, 2018). Additionally, developing dedicated Android and iOS mobile applications could improve accessibility and

user engagement, as mobile health platforms have been shown to support sustained interaction (Free et al., 2013). From a system architecture perspective, future iterations of CalorieAi should transition external API calls (such as the Gemini computer vision model) from the client-side frontend to a secure backend environment, such as Firebase Cloud Functions, to ensure enterprise-grade security for API keys (Fielding & Taylor, 2017). Furthermore, expanding the gamification engine to include social sharing and community leaderboards could further enhance long-term user retention (Lyons et al., 2014).

## **5.5 Conclusion**

This study designed and developed an AI-based web application, CalorieAi, to monitor energy expenditure and estimate daily caloric requirements. This study addressed limits in current fitness tracking systems and showed that intelligent, user-focused web applications can support health and fitness management. The system combined validated scientific models with AI-based logic to provide accurate, personalized, and actionable feedback.

Systematic testing and evaluation showed that the application, CalorieAi, worked as intended, was usable, and was acceptable to the target users (ISO 9241-11, 2018). This study adds to research on digital health technologies by proposing a practical, scalable approach to calorie tracking and weight management. These findings support further research and later improvements, and they suggest that AI-based systems like CalorieAi remain relevant for encouraging healthier lifestyle choices and improving public health outcomes (World Health Organization, 2022).

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## APPENDICES

### Appendix A: Programming Codes

landing page

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>CalorieAI | Science-Based Calorie & Macro Tracker</title>
```

```
<meta name="description"
```

```
content="Master your energy balance with CalorieAI. Precision calorie tracking, MET-based workout calculations, and AI-driven insights designed for athletes and fitness enthusiasts.">
```

```
<meta name="keywords"
```

```
content="calorie tracker, macro tracker, BMR calculator, fitness app, nutrition tracking, MET workout, CalorieAI">
```

```
<!-- Open Graph / Facebook -->
```

```
<meta property="og:type" content="website">
```

```
<meta property="og:url" content="https://calorieai.app/">
```

```
<meta property="og:title" content="CalorieAI | Master Your Energy Balance">
```

```
<meta property="og:description"
```

```
content="Precision calorie and macro tracking backed by science. Calculate your BMR and track every workout with accuracy.">
```

```
<meta property="og:image" content="images/logo-preview.png">
```

```

<!-- Twitter -->

<meta property="twitter:card" content="summary_large_image">

<meta property="twitter:url" content="https://calorieai.app/">

<meta property="twitter:title" content="CalorieAI | Master Your Energy Balance">

<meta property="twitter:description"

content="Precision calorie and macro tracking backed by science. Calculate your BMR and
track every workout with accuracy.">

<meta property="twitter:image" content="images/logo-preview.png">


<!-- Favicon -->

<link rel="icon" type="image/gif" href="images/flame-logo.gif">


<!-- Google Fonts -->

<link rel="preconnect" href="https://fonts.googleapis.com">

<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

<link

href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@300;400;500;600
;700;800&family=Space+Grotesk:wght@500;600;700&family=Orbitron:wght@400;600;70
0;900&family=Rajdhani:wght@300;400;500;600;700&family=Michroma&display=swap"

rel="stylesheet">


<!-- FontAwesome -->

<link          rel="stylesheet"          href="https://cdn.jsdelivr.net/npm/font-
awesome/6.5.1/css/all.min.css">

```

```

<!-- AOS Animation Library -->

<link href="https://unpkg.com/aos@2.3.1/dist/aos.css" rel="stylesheet">


<!-- Custom CSS -->

<link rel="stylesheet" href="css/style.css">

<link rel="stylesheet" href="css/logo.css">

</head>


<body class="antialiased">


<!-- Navigation -->

<nav class="navbar" id="navbar">

<div class="nav-container">

<div class="nav-content">

<!-- Custom Logo -->

<a href="index.html" class="logo gif-logo" style="text-decoration: none;">

<div class="logo-icon">



</div>

<div class="logo-text">

<span class="logo-brand">Calorie</span>

<span class="logo-ai">AI</span>

</div>

</a>

```

<!-- Desktop Menu -->

<div class="nav-menu">

<a href="#features" class="nav-link"><span>Feat</span><span class="nav-  
accent">ures</span></a>

<a href="#calculator" class="nav-link"><span>Calcul</span><span class="nav-  
accent">ator</span></a>

<a href="#testimonials" class="nav-link"><span>Testimon</span><span  
class="nav-accent">ial</span></a>

<a href="#how-it-works" class="nav-link"><span>How it</span> <span  
class="nav-accent">Works</span></a>

</div>

<!-- CTA -->

<div class="nav-cta">

<a href="auth.html?mode=signin" class="nav-signin-link">

<span>Sign</span><span class="nav-accent">In</span>

</a>

<a href="auth.html?mode=signup" class="nav-signup-btn">

<span>Get</span><span class="nav-accent"> Started</span> <i class="fa-solid fa-arrow-  
right"></i>

</a>

</div>

<!-- Mobile Menu Button -->

<button class="mobile-menu-btn" id="mobileMenuBtn">

<i class="fa-solid fa-bars"></i>

```

</button>

</div>

</div>

<!-- Mobile Menu -->

<div class="mobile-menu" id="mobileMenu">

<a href="#features" class="mobile-link"><span>Feat</span><span class="nav-
accent">ures</span></a>

<a href="#calculator" class="mobile-link"><span>Calcul</span><span class="nav-
accent">ator</span></a>

<a href="#testimonials" class="mobile-link"><span>Testimon</span><span class="nav-
accent">ial</span></a>

<a href="#how-it-works" class="mobile-link"><span>How it</span> <span class="nav-
accent">Works</span></a>

<a href="auth.html?mode=signin" class="mobile-link"><span>Sign</span><span
class="nav-accent">In</span></a>

</div>

</nav>

<!-- Hero Section -->

<section class="hero">

<div class="hero-bg">

<video autoplay muted loop playsinline class="hero-img">

<source src="images/E1B7AA96-E3BB-41B3-95A6-749389F62FE3.MP4"
type="video/mp4">

</video>

<div class="hero-overlay"></div>

```



</div>

<div class="hero-content">

<div class="hero-text" data-aos="fade-up" data-aos-duration="1000">

<h1 class="hero-title">

Master Your <br>

<span class="text-gradient italic">Energy Balance</span>

</h1>

<p class="hero-description">

Everything you need to track calories, manage macros, and reach your target weight with scientific

precision.

</p>

<div class="hero-buttons">

<a href="auth.html?mode=signup" class="btn-primary">

Get Started Free <i class="fa-solid fa-arrow-right"></i>

</a>

<a href="#features" class="btn-secondary">

Learn More

</a>

</div>

</div>

</div>

</section>

<!-- Science Pillars Section (Holographic Command Center) -->

<section class="science-pillars">

<div class="vignette"></div>

<!-- backgrounds -->

<div class="hologram-grid"></div>

<div class="radar-system"></div>

<div class="radar-sweep"></div>

<div class="pillars-inner">

<!-- header -->

<div class="pillars-header-hologram">

<div data-aos="fade-right">

<div class="hologram-eyebrow">

<span class="eyebrow-dot"></span>

Our Scientific Foundation

</div>

<h2>

Built on<br>

<span class="hi">Clinical Data,</span><br>

Not Guesswork.

</h2>

</div>

<div class="meta-stats" data-aos="fade-left">

<span class="meta-num">03</span>

<span class="meta-label">Active Protocols</span>

```

</div>

</div>

<!-- cards -->

<div class="hologram-cards">

<!-- Card 1 -->

<div class="hologram-card" data-aos="fade-up" data-aos-delay="100">

<div class="card-scan-line"></div>

<div class="card-corner-bracket"></div>

<div class="card-corner-bracket br"></div>

<span class="hologram-card-id">NODE-001</span>

<div class="hologram-icon-box">

<i class="fa-solid fa-calculator"></i>

</div>

<h3>Mifflin-St Jeor Equation</h3>

<div class="hologram-accent-hr"></div>

<p>The clinical gold standard for calculating Basal Metabolic Rate (BMR). We use this to
determine

your

body's exact energy needs at rest with 95% accuracy.</p>

<span class="hologram-tag">Clinical Standard</span>

<div class="calibration">

<span>LAT: 40.7128° N</span>

<span>LNG: 74.0060° W</span>

</div>

```

</div>

<!-- Card 2 (featured) -->

<div class="hologram-card featured" data-aos="fade-up" data-aos-delay="200">

<div class="card-scan-line"></div>

<div class="card-corner-bracket"></div>

<div class="card-corner-bracket br"></div>

<span class="hologram-card-id">NODE-002</span>

<div class="hologram-icon-box">

<i class="fa-solid fa-bolt"></i>

</div>

<h3>MET-Based Tracking</h3>

<div class="hologram-accent-hr"></div>

<p>Metabolic Equivalent of Task (MET) allows us to quantify the intensity of your workouts.  
We log

exertion accurately, from HIIT to resting heart rate.</p>

<span class="hologram-tag">Exercise Physiology</span>

<div class="calibration">

<span>BPM: 142 AVG</span>

<span>STATUS: LOGGING</span>

</div>

</div>

<!-- Card 3 -->

<div class="hologram-card" data-aos="fade-up" data-aos-delay="300">

<div class="card-scan-line"></div>

<div class="card-corner-bracket"></div>

<div class="card-corner-bracket br"></div>

<span class="hologram-card-id">NODE-003</span>

<div class="hologram-icon-box">

<i class="fa-solid fa-microchip"></i>

</div>

<h3>Adaptive AI Heuristics</h3>

<div class="hologram-accent-hr"></div>

<p>Our logic engine analyzes your energy balance in real-time. It suggests 20% deficits for fat loss

or

10% surpluses for lean muscle gain based on your trends.</p>

<span class="hologram-tag">Real-Time Analysis</span>

<div class="calibration">

<span>CPU: 84% LOAD</span>

<span>SYNC: ENCRYPTED</span>

</div>

</div>

</div>

</div>

</section>

<!-- Features Grid Section -->

<section id="features" class="features">

```

<div class="features-container">

<div class="features-header" data-aos="fade-right">

<span class="section-label">What you get</span>

<h2 class="section-title">Train hard. Eat right. Track <br> everything.</h2>

</div>


<div class="features-grid">

<!-- Large Workout Card -->

<a href="activity.html" class="feature-card feature-card-large" data-aos="fade-up" data-aos-
delay="100"

style="text-decoration: none;">

<div class="feature-bg workout-bg"></div>

<div class="feature-overlay"></div>

<div class="feature-content">

<div class="feature-icon">

<i class="fa-solid fa-dumbbell"></i>

</div>

<h3 class="feature-title">Workout Tracking</h3>

<p class="feature-description">Log every exercise with accurate MET-based calorie
calculations.

See exactly how much energy you burn.</p>

</div>

</a>


<!-- Right Column Stack -->

```

```

<div class="features-stack">

<!-- Nutrition Card -->

<a href="meal-history.html" class="feature-card feature-card-small" data-aos="fade-up"
data-aos-delay="200" style="text-decoration: none;">

<div class="feature-bg nutrition-bg"></div>

<div class="feature-overlay"></div>

<div class="feature-content">

<div class="feature-icon">

<i class="fa-solid fa-utensils"></i>

</div>

<h3 class="feature-title">Nutrition Logging</h3>

<p class="feature-description">Track macros, calories, and meals with a curated food
database.</p>

</div>

</a>

<!-- Goal Card -->

<a href="goal.html" class="feature-card feature-card-small" data-aos="fade-up" data-aos-
delay="300"
style="text-decoration: none;">

<div class="feature-bg goal-bg"></div>

<div class="feature-overlay"></div>

<div class="feature-content">

<div class="feature-icon">

<i class="fa-solid fa-chart-line"></i>

```

```

</div>

<h3 class="feature-title">Goal Tracking</h3>

<p class="feature-description">Set weight goals and watch your progress with detailed charts
and weekly summaries.</p>

</div>

</a>

</div>

</div>

</div>

</section>

```

```

<!-- Interactive Calculator Section -->

<section id="calculator" class="calculator">

<div class="calculator-glow"></div>

<div class="calculator-container" data-aos="zoom-in">

<div class="calculator-card">

<div class="calculator-header">

<h2 class="calculator-title">Try our BMR Calculator</h2>

<p class="calculator-subtitle">See how many calories you burn at rest instantly.</p>

</div>

<form id="bmrForm" class="calculator-form" onsubmit="calculateBMR(event)">

<div class="form-grid">

<div class="form-group">

```



```
<input type="number" id="weight" placeholder="Weight (kg)" required class="form-input">
```

```
</div>
```

```
<div class="form-group">
```

```
<input type="number" id="height" placeholder="Height (cm)" required class="form-input">
```

```
</div>
```

```
<div class="form-group">
```

```
<input type="number" id="age" placeholder="Age" required class="form-input">
```

```
</div>
```

```
<div class="form-group">
```

```
<select id="gender" class="form-select">
```

```
<option value="male">Male</option>
```

```
<option value="female">Female</option>
```

```
</select>
```

```
<div class="select-arrow">
```

```
<i class="fa-solid fa-chevron-down"></i>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<button type="submit" class="btn-calculate">
```

```
Calculate Now <i class="fa-solid fa-arrow-right"></i>
```

```
</button>
```

```
</form>
```

```
<!-- Result Display -->
```

```

<div id="resultContainer" class="result-container hidden">

<p class="result-label">Your Basal Metabolic Rate (BMR)</p>

<div class="result-value">

<span id="bmrValue">0</span> <span class="result-unit">kcal/day</span>

</div>

<p class="result-note">This is the energy your body needs to function at rest.</p>

</div>

</div>

</div>

</section>

```

```

<!-- Testimonials Section -->

<section id="testimonials" class="testimonials">

<div class="testimonials-container">

<div class="testimonials-header" data-aos="fade-up">

<span class="section-label">Testimonials</span>

<h2 class="section-title">Loved by athletes and <br>fitness enthusiasts</h2>

</div>

```

```

<div class="testimonials-grid" id="testimonialsGrid">

<!-- Testimonials will be injected here by JavaScript -->

</div>

```

```

<!-- Stats -->

<div class="testimonial-stats" data-aos="fade-up">

```

```

<div class="stat-item">

<div class="stat-number">1,200+</div>

<div class="stat-label">Active Users</div>

</div>

<div class="stat-item">

<div class="stat-number">50,000+</div>

<div class="stat-label">Meals Logged</div>

</div>

<div class="stat-item">

<div class="stat-number">99.9%</div>

<div class="stat-label">Uptime</div>

</div>

<div class="stat-item">

<div class="stat-number">4.8/5</div>

<div class="stat-label">User Satisfaction</div>

</div>

</div>

</div>

</div>

</section>

<!-- How It Works Section -->

<section id="how-it-works" class="how-it-works">

<div class="how-container">

<div class="how-content">

<div class="how-text" data-aos="fade-right">

```

<span class="section-label">How it works</span>

<h2 class="section-title">Five minutes to set up. A lifetime of results.</h2>

<div class="steps">

<div class="step">

<div class="step-number">01</div>

<div class="step-content">

<h4 class="step-title">Create your profile</h4>

<p class="step-description">Enter your measurements — we handle the math.</p>

</div>

</div>

<div class="step">

<div class="step-number">02</div>

<div class="step-content">

<h4 class="step-title">Get your targets</h4>

<p class="step-description">BMR calculated with the Mifflin-St Jeor equation.</p>

</div>

</div>

<div class="step">

<div class="step-number">03</div>

<div class="step-content">

<h4 class="step-title">Log everything</h4>

<p class="step-description">Meals, workouts, weight — all in one place.</p>

</div>

</div>

```

<div class="step">

<div class="step-number">04</div>

<div class="step-content">

<h4 class="step-title">See progress</h4>

<p class="step-description">Charts, trends, and AI insights that make sense.</p>

</div>

</div>

</div>

</div>

</div>

<div class="how-image" data-aos="fade-left">

<div class="image-wrapper">



</div>

<div class="image-glow"></div>

</div>

</div>

</div>

</div>

</section>

<!-- CTA Section -->

<section class="cta">

<div class="cta-bg">

<video autoplay muted loop playsinline class="cta-video">

```

```
<source src="images/93DFBC75-6AB4-4CD7-97FD-FDDD9E113D79.mov"
type="video/quicktime">
```

```
<source src="images/93DFBC75-6AB4-4CD7-97FD-FDDD9E113D79.mov"
type="video/mp4">
```

```
</video>
```

```
</div>
```

```
<div class="cta-overlay"></div>
```

```
<div class="cta-content" data-aos="fade-up">
```

```
<h2 class="cta-title">Stop guessing.<br>Start tracking.</h2>
```

```
<p class="cta-description">Join CalorieAI for free and get personalized, transparent calorie
tracking backed
```

```
by science.</p>
```

```
<a href="auth.html?mode=signup" class="btn-cta">
```

```
Create Free Account <i class="fa-solid fa-arrow-right"></i>
```

```
</a>
```

```
</div>
```

```
</section>
```

```
<!-- Footer -->
```

```
<footer class="footer">
```

```
<div class="footer-top">
```

```
<div class="footer-container">
```

```
<div class="footer-grid">
```

```
<!-- Brand Column -->
```

```
<div class="footer-brand">
```

```
<a href="index.html" class="logo gif-logo footer-logo" style="text-decoration: none;">
```

```

<div class="logo-icon">



</div>

<div class="logo-text">

<span class="logo-brand">Calorie</span>

<span class="logo-ai">AI</span>

</div>

</a>

<p class="footer-description">Science-based calorie tracking with complete transparency.
Master
your energy balance with precision tools designed for serious athletes and health
enthusiasts.</p>

<div class="footer-social">

<a href="https://x.com/calorieai" class="social-link" target="_blank"><i
class="fa-brands fa-x-twitter"></i></a>

<a href="https://instagram.com/calorieai" class="social-link" target="_blank"><i
class="fa-brands fa-instagram"></i></a>

<a href="https://youtube.com/@calorieai" class="social-link" target="_blank"><i
class="fa-brands fa-youtube"></i></a>

<a href="https://tiktok.com/@calorieai" class="social-link" target="_blank"><i
class="fa-brands fa-tiktok"></i></a>

</div>

</div>

<!-- Product Links -->

```

```

<div class="footer-links">

<h4>Product</h4>

<ul>

<li><a href="#features"><i class="fa-solid fa-chevron-right"></i> Features</a></li>

<li><a href="#pricing"><i class="fa-solid fa-chevron-right"></i> Pricing</a></li>

<li><a href="api.html"><i class="fa-solid fa-chevron-right"></i> API Access</a></li>

<li><a href="integrations.html"><i class="fa-solid fa-chevron-right"></i> Integrations</a>

</li>

</ul>

</div>

```

```

<!-- Resources Links -->

```

```

<div class="footer-links">

<h4>Resources</h4>

<ul>

<li><a href="blog.html"><i class="fa-solid fa-chevron-right"></i> Blog</a></li>

<li><a href="guides.html"><i class="fa-solid fa-chevron-right"></i> Guides</a></li>

<li><a href="help.html"><i class="fa-solid fa-chevron-right"></i> Help Center</a></li>

<li><a href="community.html"><i class="fa-solid fa-chevron-right"></i>
Community</a></li>

</ul>

</div>

```

```

<!-- Company Links -->

```

```

<div class="footer-links">

```



<h4>Company</h4>

<ul>

<li><a href="about.html"><i class="fa-solid fa-chevron-right"></i> About</a></li>

<li><a href="privacy.html"><i class="fa-solid fa-chevron-right"></i> Privacy Policy</a></li>

<li><a href="terms.html"><i class="fa-solid fa-chevron-right"></i> Terms of Service</a></li>

</ul>

</div>

<!-- Newsletter -->

<div class="footer-newsletter">

<h4>Stay Updated</h4>

<p>Get the latest nutrition tips and app updates delivered to your inbox.</p>

<form class="newsletter-form" onsubmit="handleNewsletter(event)">

<input type="email" placeholder="Enter your email" class="newsletter-input" required>

<button type="submit" class="newsletter-btn">Subscribe</button>

</form>

</div>

</div>

</div>

</div>

<div class="footer-bottom">

<div class="footer-bottom-content">

<p class="footer-copyright">© 2026 CalorieAI. All rights reserved. Transparent, Science-Based Calorie

Tracking.</p>

<div class="footer-legal">

<a href="#">Cookie Settings</a>

</div>

</div>

</div>

</footer>

<!-- AOS Animation Library -->

<script src="https://unpkg.com/aos@2.3.1/dist/aos.js"></script>

<!-- Embedded JavaScript -->

<script>

// Initialize Animate On Scroll

AOS.init({

once: true,

offset: 100,

duration: 800,

easing: 'ease-out-cubic',

});

// Navbar Scroll Effect

const navbar = document.getElementById('navbar');

const mobileMenuBtn = document.getElementById('mobileMenuBtn');

const mobileMenu = document.getElementById('mobileMenu');

```

window.addEventListener('scroll', () => {

if (window.scrollY > 50) {

navbar.classList.add('scrolled');

} else {

navbar.classList.remove('scrolled');

}

});

// Mobile Menu Toggle

mobileMenuBtn.addEventListener('click', () => {

mobileMenu.classList.toggle('active');

});

// Close mobile menu when clicking a link

document.querySelectorAll('.mobile-link').forEach(link => {

link.addEventListener('click', () => {

mobileMenu.classList.remove('active');

});

});

// BMR Calculator Logic (Mifflin-St Jeor Equation)

function calculateBMR(e) {

e.preventDefault();

```

```

const weight = parseFloat(document.getElementById('weight').value);

const height = parseFloat(document.getElementById('height').value);

const age = parseFloat(document.getElementById('age').value);

const gender = document.getElementById('gender').value;


if (!weight || !height || !age) return;


let bmr = 0;


// Mifflin-St Jeor Equation

if (gender === 'male') {

  bmr = (10 * weight) + (6.25 * height) - (5 * age) + 5;

} else {

  bmr = (10 * weight) + (6.25 * height) - (5 * age) - 161;

}


// Round to nearest whole number

bmr = Math.round(bmr);


// Display Result

const resultContainer = document.getElementById('resultContainer');

const bmrValue = document.getElementById('bmrValue');

resultContainer.classList.remove('hidden');

```

```

// Number animation

animateNumber(0, bmr, 1000, (val) => {

bmrValue.innerText = val;

});

// Smooth scroll to result

resultContainer.scrollToView({ behavior: 'smooth', block: 'nearest' });

}

// Newsletter Handler

function handleNewsletter(e) {

e.preventDefault();

const input = e.target.querySelector('input');

const btn = e.target.querySelector('button');

const originalText = btn.innerText;

btn.innerText = 'Subscribed!';

btn.style.background = 'var(--color-green-hover)';

input.value = "";

setTimeout(() => {

btn.innerText = originalText;

btn.style.background = "";

}, 2000);

}

```

```

// Number animation helper

function animateNumber(start, end, duration, callback) {

  const startTime = performance.now();

  function update(currentTime) {

    const elapsed = currentTime - startTime;

    const progress = Math.min(elapsed / duration, 1);

    // Ease out quart

    const ease = 1 - Math.pow(1 - progress, 4);

    const currentVal = Math.floor(start + (end - start) * ease);

    callback(currentVal);

    if (progress < 1) {
      requestAnimationFrame(update);
    }
  }

  requestAnimationFrame(update);
}

// Testimonials Pool (9 items)

const testimonialsPool = [

```

```

{
  name: "Marcus Jordan",
  role: "Professional Bodybuilder",
  image: "https://images.unsplash.com/photo-1594381898411-846e7d193883?q=80&w=200",
  text: "CalorieAI completely changed how I approach my nutrition. The BMR calculator is spot-on, and I've lost 15kg in 4 months while maintaining muscle mass.",
  stars: 5
},
{
  name: "Sarah Williams",
  role: "Marathon Runner",
  image: "https://images.unsplash.com/photo-1552674605-db6ffd4facb5?q=80&w=200",
  text: "As a marathon runner, I need precise calorie tracking. The MET-based workout calculations are incredibly accurate. Optimized my fueling strategy!",
  stars: 5
},
{
  name: "David Adeyemi",
  role: "CrossFit Coach",
  image: "https://images.unsplash.com/photo-1522512115668-c09775d6f424?q=80&w=200",
  text: "Finally, a fitness app that doesn't feel like a black box. Seeing the actual formulas used helped me understand my body better. Intuitive and beautiful.",
  stars: 5
},
{
  name: "Nia Robinson",

```

role: "Yoga Instructor",

image: "https://images.unsplash.com/photo-1599447421416-3414500d18a5?q=80&w=200",

text: "The mindfulness of tracking my energy balance has improved my practice significantly. CalorieAI is the most elegant tool I've used for health.",

stars: 5

},

{

name: "Tariq Smith",

role: "HIIT Specialist",

image: "https://images.unsplash.com/photo-1590556409324-aa1d726e5c3c?q=80&w=200",

text: "I push my clients to use this because transparency matters. No guesswork, just results based on real science. The AI insights are a game changer.",

stars: 5

},

{

name: "Aaliyah Banks",

role: "Powerlifter",

image: "https://images.unsplash.com/photo-1541534741688-6078c6bfb5c5?q=80&w=200",

text: "Precision is everything when you're training for a meet. CalorieAI gives me the data I need to stay in my weight class while crushing PRs.",

stars: 5

},

{

name: "Kofi Owusu",

role: "Personal Trainer",

image: "https://images.unsplash.com/photo-1622343106395-585353841961?q=80&w=200",



text: "The best part is the community and the science. My clients love the dashboard and it makes my job so much easier to see their real-time progress.",

stars: 5

},

{

name: "Maya Evans",

role: "Tennis Player",

image: "https://images.unsplash.com/photo-1595301758535-618476a6663f?q=80&w=200",

text: "The performance tracking is incredible. I can see exactly how my diet affects my endurance on the court. Highly recommended for any athlete.",

stars: 4.5

},

{

name: "Jordan Okafor",

role: "Basketball Player",

image: "https://images.unsplash.com/photo-1546519638-68e109498ffc?q=80&w=200",

text: "Finally found an app that understands high-intensity sports energy expenditure correctly. CalorieAI keeps me at peak performance all season long.",

stars: 5

}

];

```
function renderRandomTestimonials() {
```

```
const grid = document.getElementById('testimonialsGrid');
```

```
if (!grid) return;
```

```

// Shuffle and pick 3

const shuffled = [...testimonialsPool].sort(() => 0.5 - Math.random());

const selected = shuffled.slice(0, 3);

grid.innerHTML = selected.map((t, index) => `
<div class="testimonial-card" data-aos="fade-up" data-aos-delay="${(index + 1) * 100}">
<div class="testimonial-stars">

${Array(Math.floor(t.stars)).fill('<i class="fa-solid fa-star"></i>').join("")}

${t.stars % 1 !== 0 ? '<i class="fa-solid fa-star-half-stroke"></i>' : ""}

</div>

<p class="testimonial-text">"${t.text}"</p>

<div class="testimonial-author">



<div class="author-info">

<h4>${t.name}</h4>

<p>${t.role}</p>

</div>

</div>

</div>

</div>

`).join("");
}

// Call on load

renderRandomTestimonials();

```

```

// Refresh AOS after dynamic content

AOS.refresh();


// Smooth scroll for anchor links

document.querySelectorAll('a[href^="#"]').forEach(anchor => {

anchor.addEventListener('click', function (e) {

e.preventDefault();

const target = document.querySelector(this.getAttribute('href'));

if (target) {

target.scrollIntoView({

behavior: 'smooth',

block: 'start'

});

}

});

});

</script>

<script type="module">

import { initializeApp } from "https://www.gstatic.com/firebasejs/10.7.1/firebase-app.js";

import { getAuth, onAuthStateChanged } from "https://www.gstatic.com/firebasejs/10.7.1/firebase-auth.js";

import { getFirestore, doc, getDoc } from "https://www.gstatic.com/firebasejs/10.7.1/firebase-firestore.js";

const firebaseConfig = {

```

```

apiKey: "AlzaSyDWjpdPmM0gEWByJjGZ5ALkxBpUd5n7F_g",
authDomain: "calorieai-c630a.firebaseio.com",
projectId: "calorieai-c630a",
storageBucket: "calorieai-c630a.firebaseio.com",
messagingSenderId: "610311761992",
appId: "1:610311761992:web:83389f99535950bbfc287d",
measurementId: "G-8YJZM8WWBZ"
};

```

```
const app = initializeApp(firebaseConfig);
```

```
const auth = getAuth(app);
```

```
const db = getFirestore(app);
```

```
onAuthStateChanged(auth, async (user) => {
```

```
  if (user) {
```

```
    const signinLink = document.querySelector('.nav-signin-link');
```

```
    const navSignupBtn = document.querySelector('.btn-nav');
```

```
    const heroSignupBtn = document.querySelector('.hero-buttons .btn-primary');
```

```
    const mobileSignin = document.querySelector('.mobile-link[href*="signin"]');
```

```
    try {
```

```
      const userDoc = await getDoc(doc(db, "users", user.uid));
```

```
      const hasProfile = userDoc.exists();
```

```
      const targetUrl = hasProfile ? 'dashboard.html' : 'onboarding.html';
```

```
      const label = hasProfile ? 'Go to Dashboard' : 'Continue Setup';
```

```

// Update Nav Sign In link (hide or change)

if (signinLink) {

  signinLink.style.display = 'none';

}


// Update Nav Primary Button

if (navSignupBtn) {

  navSignupBtn.href = targetUrl;

  // Split label for consistency if possible, otherwise use full label

  const words = label.split(' ');

  if (words.length >= 2) {

    navSignupBtn.innerHTML = `<span>${words[0]}</span><span class="nav-accent">
    ${words.slice(1).join(' ')}</span> <i class="fa-solid fa-arrow-right"></i>`;

  } else {

    navSignupBtn.innerHTML = `<span>${label}</span> <i class="fa-solid fa-arrow-
    right"></i>`;

  }

}


// Update Hero Primary Button

if (heroSignupBtn) {

  heroSignupBtn.href = targetUrl;

  heroSignupBtn.innerHTML = `${label} <i class="fa-solid fa-arrow-right"></i>`;

}

```

```
if (mobileSignin) {  
  mobileSignin.href = targetUrl;  
  mobileSignin.innerText = label;  
}  
} catch (e) {  
  console.error("Auth helper error:", e);  
}  
}  
});  
</script>  
</body>  
  
</html>
```